

25.0 PEMODELAN STRUKTUR FONDASI DANGKAL - PIPE SUPPORT TYPE -1C**25.1 Gambaran Umum**

Struktur Fondasi Dangkal - *Pipe Support* Type -1C merupakan model support baru yang diterapkan pada PS-007A s.d. PS-008.

Elevasi Pipa	BOP	=	2.500	m
Kedalaman <i>Fondasi</i>	H _{Fondasi}	=	0.50	m
Total Tinggi Struktur pada Model STAAD	H	=	2.70	m

Tipe Struktur	Panjang		
	L (m)	L1 (m)	L2 (m)
<i>Pipe Support Structure</i>	1.00	0.50	0.50
	Tinggi		
	H (m)	H _{Fondasi} (m)	H1 (m)
	2.50	0.50	2.500

Dimensi Pedestal

Panjang penampang pedestal	L_p	=	400.00	mm
Lebar penampang pedestal	B_p	=	400.00	mm
Tinggi pedestal di atas permukaan tanah	$H_{p,t}$	=	300.00	mm
Tinggi pedestal di bawah permukaan tanah	$H_{p,b}$	=	200.00	mm
Tinggi total pedestal	H_p	=	500.00	mm
Luas penampang pedestal	A_p	=	$L_p \cdot B_p$	= 0.16 m ²

Dimensi Footing

Panjang minimum footing	$L_{f(min)}$	=	L_p	= 400.00 mm
Panjang footing	L_f	=	1500.00	mm (OK)
Lebar minimum footing	$B_{f(min)}$	=	B_p	= 400.00 mm
Lebar footing	B_f	=	1500.00	mm (OK)
Tinggi footing	H_f	=	300.00	mm
Luas penampang footing	A_f	=	$L_f \cdot B_f$	= 2.25 m ²
Modulus penampang elastis terhadap sb X	$S_{X,f}$	=	$(B_f \cdot L_f^2)/6$	= 0.56 m ²
Modulus penampang elastis terhadap sb Z	$S_{Z,f}$	=	$(L_f \cdot B_f^2)/6$	= 0.56 m ²
Tinggi total struktur	H_{p+f}	=	$H_p + H_f$	= 0.80 m

Berat Pedestal

Berat jenis beton,	γ_c	=	24.00	kN/m ³
Berat pedestal,	W_p	=	$A_p \cdot H_p \cdot \gamma_c$	= 3.84 kN

Berat Footing

Berat jenis beton,	γ_c	=	24.00	kN/m ³
Berat footing,	W_f	=	$A_f \cdot H_f \cdot \gamma_c$	= 16.20 kN

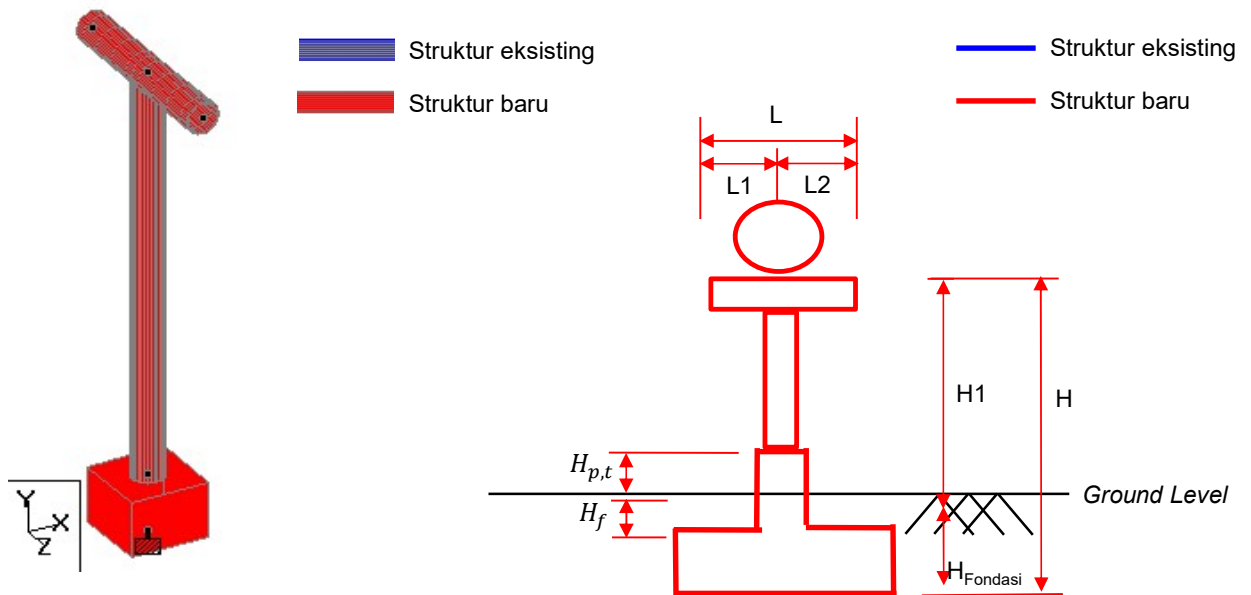
Berat Tanah di Atas Telapak

Berat jenis tanah,	γ_s	=	12.00	kN/m ³
Berat tanah di atas footing,	$W_{s,b}$	=	$(A_f - A_p) \cdot H_{p,b} \cdot \gamma_s$	= 5.02 kN

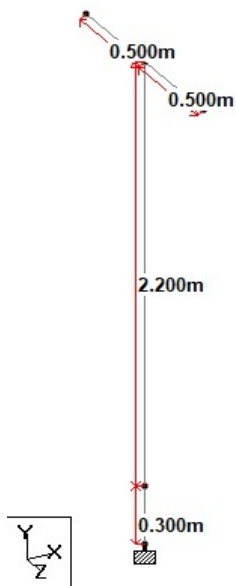
Beban Mati Total

Total beban mati pada footing,	$F_{Y,Df}$	=	$W_p + W_f + W_{s,b}$	= 25.06 kN
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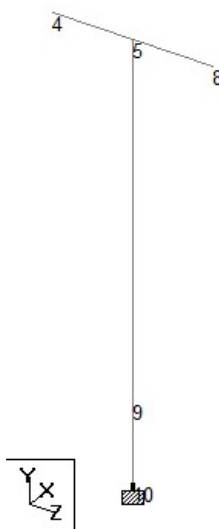
25.2 Dimensi & Pemodelan Struktur



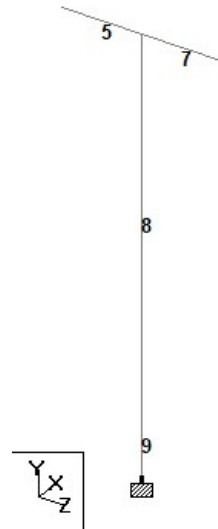
Pemodelan 3D di STAAD.Pro



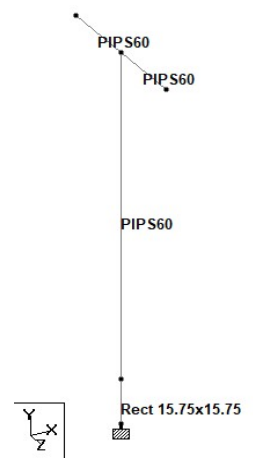
Pemodelan di STAAD Pro



Penomoran Node



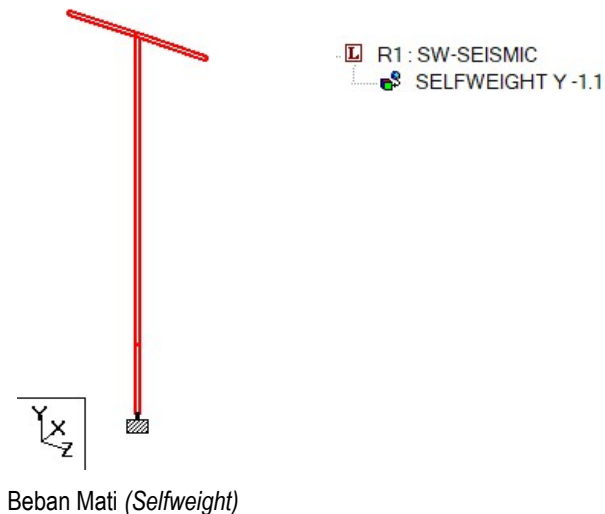
Penomoran Member



Member Properties

26.0 BEBAN RENCANA FONDASI DANGKAL - PIPE SUPPORT TYPE-1C**26.1 Beban Mati (DL)**

Beban sendiri bangunan dihitung menggunakan fitur SELFWEIGHT di STAAD Pro dengan faktor $F_y -1.1$.

**26.2 Beban Pipa (P)**

Beban pipa ditentukan berdasarkan hasil pemodelan CAESAR dengan mengambil nilai beban maksimum yang menyebabkan kondisi kritis. Beban pipa yang tidak dimodelkan dengan CAESAR ditentukan berdasarkan perhitungan berat pipa dan konten air (kondisi test) yang dihitung dengan mengacu pada ukuran pipa dan panjang bentang pipa pada data layout piping dengan schedule STD40.

- Pipa Baru (1 NOS) Diameter = 20 " = 508.00 mm

*Pemodelan untuk perhitungan beban diambil pada jenis dan tipe yang paling kritis.

*Input beban di tengah pipa menggunakan metode beban terpusat pada member (Beban/2)

No.	Tipe Pipe Support	Nomor Pipe Support	Span Max. (m)	Diameter Pipa (in)	Beban Pipa (kN)		
					Empty (FY)	Operating (FY)	Test (FY)
1	Type-1C	PS-007A s.d. PS-008.	6.50	20.00	7.47	27.04	28.93

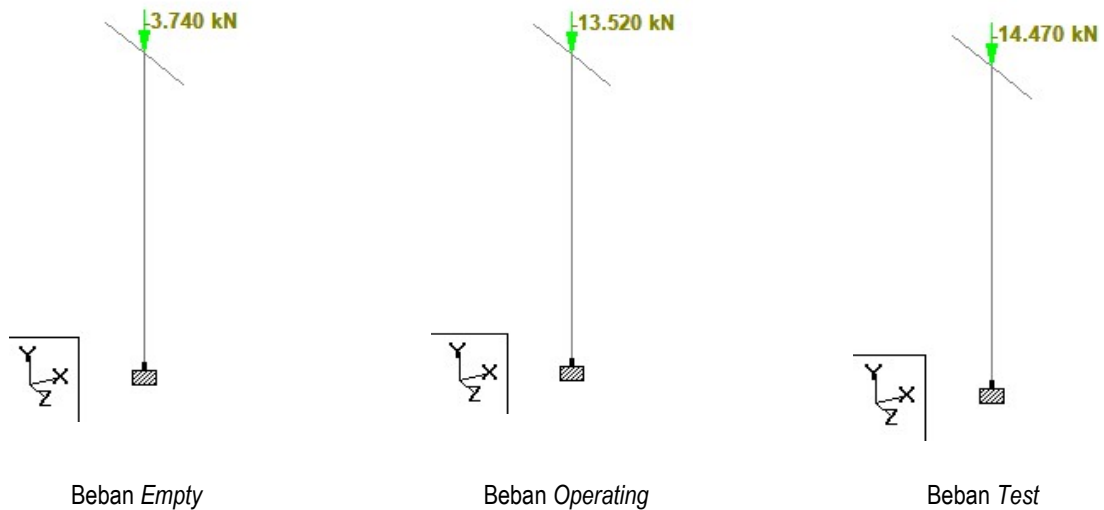
26.3 Beban Termal Pipa (T)

Beban termal terdiri atas beban termal friction dan termal expansion, terdapat tiga kondisi untuk menentukan besarnya beban **Thermal Friction (FX)** yaitu:

1. Untuk pipa baru, diperoleh dari *pipe stress* (CAESAR)
2. 30% dari berat *operation* pipa terbesar
3. 7.5% dari total berat *operation* semua pipa (baru + eksisting)

Beban **Thermal Expansion (FZ)** diperoleh dari data CAESAR

No.	Tipe Pipe Support	Nomor Pipe Support	Faktor Pengali		Beban Termal (kN)		
					(FX)	Beban Termal (FX)	Beban Termal (FZ)
1	Type-1C	PS-007A s.d. PS-008.	7.5%	Total Load		6.77	4.79
			30%	Max. Diameter			
			New Pipe (CAESAR)		6.77		



26.4 Beban Angin (W)

Beban Angin Struktur

Input Data :

Kecepatan Angin	V = 71.58 mph	(ref : PHR-SP-CI-GG-002)
Tipe Struktur	= Other structure	
Tipe Bangunan	= Open	

Kecepatan tekanan angin yang terjadi dapat ditentukan melalui persamaan berikut:

$$q_h = 0.613 * K_z * K_{zt} * K_d * K_e * V^2 \quad (\text{N/m}^2) \quad (\text{ASCE 7-16 Eq. 26.10-1 si.})$$

Desc:	K_z	= Koefisien Exposure	= 0.85	(ASCE 7-22 Tabel 26.10-1, untuk tinggi 0 - 4.6 m)
	K_{zt}	= Faktor Topografi,	= 1.00	(ASCE 7-22 Sec. 26.8.2)
	K_d	= Faktor Arah Angin	= 0.85	(ASCE 7-22 Table 26.6-1)
	K_e	= Faktor Elevasi Ground	= 1.00	(ASCE 7-22 Sec. 26.9, elevasi ground 50 m dpl*)
	V	= Kecepatan Angin	= 32.00 m/s	(ref : PHR-SP-CI-GG-002)

*dpl : di atas permukaan laut

Kecepatan Tekanan:

$$q_h = 0.613 * K_z * K_{zt} * K_d * K_e * V^2 = 453 \text{ N/m}^2 = 0.45 \text{ kN/m}^2$$

Perhitungan Beban Angin untuk Other Buildings :

$$P = q_h * G * C_f \quad (\text{N/m}^2) \quad (\text{ASCE 7-22 Sec. 29.4-1})$$

Desc. :	q_h	= Kecepatan Tekanan Angin	= 0.45 kN/m ²	
	G	= Gust Factor	= 0.85	(ASCE 7-22 Sec. 26.11)
	C_f	= Force Coefficient	= 0.65	(ASCE 7-22 Fig. 29.4-1)

Force Coefficient (C_f)

Cross Section	h/D	C_f
Round, $D\sqrt{qz} > 2.5$	15.24	0.65

$$\begin{aligned} \text{Beban Angin Rencana } P &= q_h * G * C_f = 0.25 \text{ kN/m}^2 \\ \text{Beban Angin Minimum } P_{\min} &= 0.77 \text{ kN/m}^2 \\ \text{Beban Angin yang Digunakan } P_{\text{use}} &= 0.77 \text{ kN/m}^2 \end{aligned}$$

Beban angin pada pipa ditransfer ke beam sebagai beban titik:

Dia. Pipa (in)	Arah	Lebar (m)	P_{use} (kN/m ²)	Span (m)	A_f (m ²)	W_x (kN)	W_z (kN)
20.00	Z-dir.	0.254	0.77	6.50	1.65	0.00	1.27

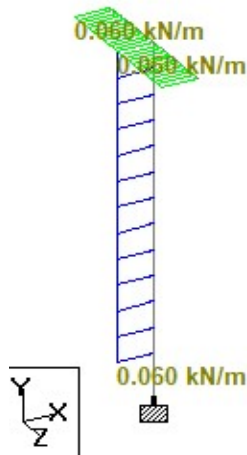
Beban angin pada struktur diberikan sebagai beban terdistribusi:

X-direction

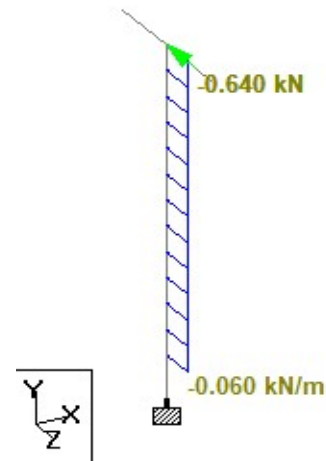
Member	Lebar (m)	P_{use} (kN/m ²)	W_x (kN)
Beam PIPS60	0.084	0.77	0.06
Beam PIPS60	0.084	0.77	0.06

Z-direction

Member	Lebar (m)	P_{use} (kN/m ²)	W_z (kN)
Beam PIPS60	0.084	0.77	0.06



Beban Angin Arah X



Beban Angin Arah Z

26.5 Beban Gempa (V)

Penentuan parameter beban gempa pada pipe support dilakukan berdasarkan lokasi proyek:

Lokasi = DURI CGS 1

Latitude = 1.28647

Longitude = 101.224779

Sumber: <http://rsa.ciptakarya.pu.go.id/2021>

Parameter Gempa	Simbol	Nilai	Satuan
Klasifikasi tanah lokasi, tanah sedang ($15 < N < 50$)	Site	SE	-
Percepatan puncak tanah (PGA) pada peta MCE _R	PGA	0.184	g
Parameter percepatan respons spektral MCE _R terpetakan pada periode 0,2 detik	S _s	0.368	g
Parameter percepatan respons spektral MCE _R terpetakan pada periode 1 detik	S ₁	0.256	g
Parameter percepatan respons spektral desain pada periode 0,2 detik	S _{DS}	0.521	g
Parameter percepatan respons spektral desain pada periode 1 detik	S _{D1}	0.508	g
Koefisien lokasi untuk periode pendek	Fa	2.122	g
Koefisien lokasi untuk periode panjang	Fv	2.977	g

Kategori Desain Gempa	Simbol	Nilai	Satuan
Kategori Risiko Bangunan	RC	III	-
Faktor Keutamaan (I)	I _e	1.25	-
Faktor Reduksi Gempa (R)	R	2.5	-
Faktor Perbesaran Lendutan (Cd)	Cd	2.5	-
Faktor Redundansi (ρ)	ρ	1.3	-

(Table 3, SNI-1726-2019)

(Table 4, SNI-1726-2019)

(Table 28, SNI-1726-2019)

(Section 7.3.4, SNI-1726-2019)

Tabel 28 – Koefisien seismik untuk struktur nongedung serupa gedung

Jenis struktur bangunan nongedung	Persyaratan Pendetailan	R	Ω ₀	C _d	Batasan Sistem Struktur dan Batasan Tinggi Bangunan, h _n , (m) ^a				
					Kategori Desain Seismik				
					B	C	D ^b	E ^b	F ^c
Rangka baja pemikul momen biasa	AISC 341	3,5	3	3	TB	TB	T ⁽¹⁾ ₃₀	T ⁽¹⁾ ₃₀	T ⁽¹⁾ ₃₀
• Dengan tambahan ketinggian yang diizinkan	AISC 341	2,5	2	2,5	TB	TB	30	30	T ⁽¹⁾ ₃₀
Tanpa batasan ketinggian	AISC 360	1	1	1	TB	TB	TB	TB	TB
Rangka beton bertulang pemikul momen biasa	SNI 2847, tidak termasuk pasal 0	3	3	2,5	TB	TI	TI	TI	TI
• Dengan tambahan ketinggian yang diizinkan	SNI 2847, tidak termasuk pasal 0	0,8	1	1	TB	TB	15	15	15

Respons Spektrum Desain

$$S_{DS} = 0.521$$

$$S_{D1} = 0.508$$

$$T_0 = 0.2 * S_{D1} / S_{DS} = 0.2 * 0.51 / 0.52 = 0.195 \text{ s}$$

$$T_s = S_{D1} / S_{DS} = 0.51 / 0.52 = 0.975 \text{ s}$$

Perhitungan Koefisien Respons Seismik

$$\text{Koefisien } C_u = 1.40 \quad (\text{SNI-1726-2019, Table 17})$$

$$\text{Koefisien } x = 0.75 \quad (\text{SNI-1726-2019, Table 18})$$

$$\text{Koefisien periode bangunan } C_t = 0.0724 \quad (\text{SNI-1726-2019, Table 18})$$

$$\text{Tinggi bangunan } h_n = 2.70 \text{ m}$$

$$\text{Periode fundamental perkiraan } T_a = C_t * h_n^x = 0.152 \text{ s} \quad (\text{SNI-1726-2019, Eq. 36})$$

$$\text{Faktor modifikasi respons (R) x-dir } R_x = 2.50 \quad (\text{Design spec./SNI-1726-2019})$$

$$\text{z-dir } R_z = 2.50 \quad (\text{Design spec./SNI-1726-2019})$$

$$\text{Periode struktur berdasarkan hasil analisis STAAD } T_c = 0.300 \text{ s} \quad (\text{output dari STAAD})$$

$$\text{Periode struktur maksimum } T_{max} = C_u * T_a = 0.213 \text{ s}$$

$$\text{Periode struktur yang digunakan } T_{use} = 0.213 \text{ s}$$

$$\text{if } T_c < T_a \rightarrow T_{use} = T_a$$

$$\text{if } T_a < T_c < T_{max} \rightarrow T_{use} = T_c$$

$$\text{if } T_c > T_{max} \rightarrow T_{use} = T_{max}$$

Respon Spektrum Desain

$$\begin{aligned}
 \text{for } T < T_o & \rightarrow S_a = S_{DS} & = 0.521 & \text{digunakan } S_a = 0.521 \\
 \text{for } T \leq T_s & \rightarrow S_a = S_{DS} & = 0.521 & \\
 \text{for } T > T_s & \rightarrow S_a = S_{D1} / T & = 2.379 & \\
 C_s & = S_a * I_e / R & = 0.2605 & \\
 C_{s \min} & = \max(0.044 * S_{DS} * I_e; 0.01) & = 0.0287 & \\
 C_{s \max} & = S_{D1} / (T * (R / I_e)) & = 1.1897 & \\
 \hline
 & & \mathbf{C_s = 0.261} &
 \end{aligned}$$

Output STAAD.Pro :

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*****
* IBC 2015 SEISMIC LOAD ALONG X :
* CT = 0.072 Cu = 1.400 x = 0.8000
* TIME PERIODS :
* Ta = 0.120 T = 0.045 Tuser = 0.000
* TIME PERIOD USED (T) = 0.045
* Cs LIMITS : LOWER = 0.029 UPPER = 5.610
* LOAD FACTOR = 1.000
* DESIGN BASE SHEAR = 1.000 X 0.260 X 0.74
*                      = 0.19 KN
*****
* IBC 2015 SEISMIC LOAD ALONG Z :
* CT = 0.072 Cu = 1.400 x = 0.8000
* TIME PERIODS :
* Ta = 0.120 T = 0.045 Tuser = 0.000
* TIME PERIOD USED (T) = 0.045
* Cs LIMITS : LOWER = 0.029 UPPER = 5.612
* LOAD FACTOR = 1.000
* DESIGN BASE SHEAR = 1.000 X 0.260 X 0.74
*                      = 0.19 KN
*****

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26.6 Kombinasi Pembebanan

Berikut merupakan penyesuaian *load combination* terhadap $p = 1.3$ dan beban gempa vertikal.

ASD Load Comb untuk $r = 1.3$

$$\begin{aligned}
 E_h & = r & = 1.3 \\
 E_v & = 0.2 S_{DS} D & = 0.1042 D \\
 1D + 0.7E_v + 0.7E_h & & (\text{AISC 7-16 Sec. 2.4.5}) \\
 1 D & + 0.07 D & + 0.91 V \\
 1.07 D & + 0.91 V \\
 \\
 1D + 0.525E_v + 0.525E_h & & (\text{AISC 7-16 Sec. 2.4.5}) \\
 1 D & + 0.05 D & + 0.68 V \\
 1.05 D & + 0.68 V \\
 \\
 0.6D - 0.7E_v + 0.7E_h & & (\text{AISC 7-16 Sec. 2.4.5}) \\
 0.6 D & - 0.07 D & + 0.91 V \\
 0.53 D & + 0.91 V
 \end{aligned}$$

LRFD Load Comb untuk $r = 1.3$

$$\begin{aligned}
 1.2D + E_v + E_h & & (\text{AISC 7-16 Sec. 2.3.6}) \\
 1.2 D & + 0.10 D & + 1.3 V \\
 1.30 D & + 1.3 V \\
 \\
 0.9D - E_v + E_h & & (\text{AISC 7-16 Sec. 2.3.6}) \\
 0.9 D & - 0.10 D & + 1.3 V \\
 0.80 D & + 1.3 V
 \end{aligned}$$

Kombinasi beban tanpa faktor (*unfactored load combination* / ASD) untuk pemeriksaan lendutan, perpindahan, stabilitas fondasi, dan kapasitas daya dukung tanah adalah sebagai berikut:

L/C	KOMBINASI PEMBEBANAN
101	1DL+1PE+1QE+1CDL+1H
102	1DL+1PE+1QE+1CDL+1H+0.6WX

103	1DL+1PE+1QE+1CDL+1H-0.6WX
104	1DL+1PE+1QE+1CDL+1H+0.6WZ
105	1DL+1PE+1QE+1CDL+1H-0.6WZ
106	(1 + 0.14Sds)DL+(1 + 0.14Sds)PE+(1 + 0.14Sds)QE+(1 + 0.14Sds)CDL+1H+0.91VX+0.27VZ
107	(1 + 0.14Sds)DL+(1 + 0.14Sds)PE+(1 + 0.14Sds)QE+(1 + 0.14Sds)CDL+1H+0.91VX-0.27VZ
108	(1 + 0.14Sds)DL+(1 + 0.14Sds)PE+(1 + 0.14Sds)QE+(1 + 0.14Sds)CDL+1H-0.91VX+0.27VZ
109	(1 + 0.14Sds)DL+(1 + 0.14Sds)PE+(1 + 0.14Sds)QE+(1 + 0.14Sds)CDL+1H-0.91VX-0.27VZ
110	(1 + 0.14Sds)DL+(1 + 0.14Sds)PE+(1 + 0.14Sds)QE+(1 + 0.14Sds)CDL+1H+0.27VX+0.91VZ
111	(1 + 0.14Sds)DL+(1 + 0.14Sds)PE+(1 + 0.14Sds)QE+(1 + 0.14Sds)CDL+1H+0.27VX-0.91VZ
112	(1 + 0.14Sds)DL+(1 + 0.14Sds)PE+(1 + 0.14Sds)QE+(1 + 0.14Sds)CDL+1H-0.27VX+0.91VZ
113	(1 + 0.14Sds)DL+(1 + 0.14Sds)PE+(1 + 0.14Sds)QE+(1 + 0.14Sds)CDL+1H-0.27VX-0.91VZ
114	0.6DL+0.6PE+0.6QE+0.6CDL+1H+0.6WX
115	0.6DL+0.6PE+0.6QE+0.6CDL+1H-0.6WX
116	0.6DL+0.6PE+0.6QE+0.6CDL+1H+0.6WZ
117	0.6DL+0.6PE+0.6QE+0.6CDL+1H-0.6WZ
118	1DL+1PO+1QO+1TE+1TF+1CDL+1H
119	1DL+1PO+1QO+1TE+1TF+1CDL+1H+0.6WX
120	1DL+1PO+1QO+1TE+1TF+1CDL+1H-0.6WX
121	1DL+1PO+1QO+1TE+1TF+1CDL+1H+0.6WZ
122	1DL+1PO+1QO+1TE+1TF+1CDL+1H-0.6WZ
123	(1 + 0.14Sds)DL+(1 + 0.14Sds)PO+(1 + 0.14Sds)QO+(1 + 0.14Sds)TE+(1 + 0.14Sds)TF+(1 + 0.14Sds)CDL+1H+0.91VX+0.27VZ
124	(1 + 0.14Sds)DL+(1 + 0.14Sds)PO+(1 + 0.14Sds)QO+(1 + 0.14Sds)TE+(1 + 0.14Sds)TF+(1 + 0.14Sds)CDL+1H+0.91VX-0.27VZ
125	(1 + 0.14Sds)DL+(1 + 0.14Sds)PO+(1 + 0.14Sds)QO+(1 + 0.14Sds)TE+(1 + 0.14Sds)TF+(1 + 0.14Sds)CDL+1H-0.91VX+0.27VZ
126	(1 + 0.14Sds)DL+(1 + 0.14Sds)PO+(1 + 0.14Sds)QO+(1 + 0.14Sds)TE+(1 + 0.14Sds)TF+(1 + 0.14Sds)CDL+1H-0.91VX-0.27VZ
127	(1 + 0.14Sds)DL+(1 + 0.14Sds)PO+(1 + 0.14Sds)QO+(1 + 0.14Sds)TE+(1 + 0.14Sds)TF+(1 + 0.14Sds)CDL+1H+0.27VX+0.91VZ
128	(1 + 0.14Sds)DL+(1 + 0.14Sds)PO+(1 + 0.14Sds)QO+(1 + 0.14Sds)TE+(1 + 0.14Sds)TF+(1 + 0.14Sds)CDL+1H+0.27VX-0.91VZ
129	(1 + 0.14Sds)DL+(1 + 0.14Sds)PO+(1 + 0.14Sds)QO+(1 + 0.14Sds)TE+(1 + 0.14Sds)TF+(1 + 0.14Sds)CDL+1H-0.27VX+0.91VZ
130	(1 + 0.14Sds)DL+(1 + 0.14Sds)PO+(1 + 0.14Sds)QO+(1 + 0.14Sds)TE+(1 + 0.14Sds)TF+(1 + 0.14Sds)CDL+1H-0.27VX-0.91VZ
131	1DL+1PO+1QO+1TE+1TF+1LL+1CDL+1CLL+1I+1H
132	1DL+1PO+1QO+1TE+1TF+1LR+1CDL+1CLL+1I+1H
133	1DL+1PO+1QO+1TE+1TF+0.75LL+0.75LR+1CDL+1CLL+1I+1H+0.45WX
134	1DL+1PO+1QO+1TE+1TF+0.75LL+0.75LR+1CDL+1CLL+1I+1H-0.45WX
135	1DL+1PO+1QO+1TE+1TF+0.75LL+0.75LR+1CDL+1CLL+1I+1H+0.45WZ
136	1DL+1PO+1QO+1TE+1TF+0.75LL+0.75LR+1CDL+1CLL+1I+1H-0.45WZ
137	(1+0.105Sds)DL+(1+0.105Sds)PO+(1+0.105Sds)QO+(1+0.105Sds)TE+(1+0.105Sds)TF+0.75LL+(1+0.105Sds)CDL+0.75CLL+0.75I+1H+0.68VX+0.2VZ
138	(1+0.105Sds)DL+(1+0.105Sds)PO+(1+0.105Sds)QO+(1+0.105Sds)TE+(1+0.105Sds)TF+0.75LL+(1+0.105Sds)CDL+0.75CLL+0.75I+1H+0.68VX-0.2VZ
139	(1+0.105Sds)DL+(1+0.105Sds)PO+(1+0.105Sds)QO+(1+0.105Sds)TE+(1+0.105Sds)TF+0.75LL+(1+0.105Sds)CDL+0.75CLL+0.75I+1H-0.68VX+0.2VZ

140	$(1+0.105Sds)DL+(1+0.105Sds)PO+(1+0.105Sds)QO+(1+0.105Sds)TE+(1+0.105Sds)TF+0.75LL+(1+0.105Sds)CDL+0.75CLL+0.75I+1H-0.68VX-0.2VZ$
141	$(1+0.105Sds)DL+(1+0.105Sds)PO+(1+0.105Sds)QO+(1+0.105Sds)TE+(1+0.105Sds)TF+0.75LL+(1+0.105Sds)CDL+0.75CLL+0.75I+1H+0.2VX+0.68VZ$
142	$(1+0.105Sds)DL+(1+0.105Sds)PO+(1+0.105Sds)QO+(1+0.105Sds)TE+(1+0.105Sds)TF+0.75LL+(1+0.105Sds)CDL+0.75CLL+0.75I+1H+0.2VX-0.68VZ$
143	$(1+0.105Sds)DL+(1+0.105Sds)PO+(1+0.105Sds)QO+(1+0.105Sds)TE+(1+0.105Sds)TF+0.75LL+(1+0.105Sds)CDL+0.75CLL+0.75I+1H-0.2VX+0.68VZ$
144	$(1+0.105Sds)DL+(1+0.105Sds)PO+(1+0.105Sds)QO+(1+0.105Sds)TE+(1+0.105Sds)TF+0.75LL+(1+0.105Sds)CDL+0.75CLL+0.75I+1H-0.2VX-0.68VZ$
145	$0.6DL+0.6PO+0.6QO+0.6TE+0.6TF+0.6CDL+0.6WX$
146	$0.6DL+0.6PO+0.6QO+0.6TE+0.6TF+0.6CDL-0.6WX$
147	$0.6DL+0.6PO+0.6QO+0.6TE+0.6TF+0.6CDL+0.6WZ$
148	$0.6DL+0.6PO+0.6QO+0.6TE+0.6TF+0.6CDL-0.6WZ$
149	$(0.6-0.14Sds)DL+(0.6-0.14Sds)PO+(0.6-0.14Sds)QO+(0.6-0.14Sds)TE+(0.6-0.14Sds)TF+(0.6-0.14Sds)CDL+1H+0.91VX-0.27VZ$
150	$(0.6-0.14Sds)DL+(0.6-0.14Sds)PO+(0.6-0.14Sds)QO+(0.6-0.14Sds)TE+(0.6-0.14Sds)TF+(0.6-0.14Sds)CDL+1H+0.91VX+0.27VZ$
151	$(0.6-0.14Sds)DL+(0.6-0.14Sds)PO+(0.6-0.14Sds)QO+(0.6-0.14Sds)TE+(0.6-0.14Sds)TF+(0.6-0.14Sds)CDL+1H-0.91VX+0.27VZ$
152	$(0.6-0.14Sds)DL+(0.6-0.14Sds)PO+(0.6-0.14Sds)QO+(0.6-0.14Sds)TE+(0.6-0.14Sds)TF+(0.6-0.14Sds)CDL+1H-0.91VX-0.27VZ$
153	$(0.6-0.14Sds)DL+(0.6-0.14Sds)PO+(0.6-0.14Sds)QO+(0.6-0.14Sds)TE+(0.6-0.14Sds)TF+(0.6-0.14Sds)CDL+1H+0.27VX+0.91VZ$
154	$(0.6-0.14Sds)DL+(0.6-0.14Sds)PO+(0.6-0.14Sds)QO+(0.6-0.14Sds)TE+(0.6-0.14Sds)TF+(0.6-0.14Sds)CDL+1H+0.27VX-0.91VZ$
155	$(0.6-0.14Sds)DL+(0.6-0.14Sds)PO+(0.6-0.14Sds)QO+(0.6-0.14Sds)TE+(0.6-0.14Sds)TF+(0.6-0.14Sds)CDL+1H-0.27VX+0.91VZ$
156	$(0.6-0.14Sds)DL+(0.6-0.14Sds)PO+(0.6-0.14Sds)QO+(0.6-0.14Sds)TE+(0.6-0.14Sds)TF+(0.6-0.14Sds)CDL+1H-0.27VX-0.91VZ$
157	$1DL+1PT+1QT+1CDL+1H+0.45WX$
158	$1DL+1PT+1QT+1CDL+1H-0.45WX$
159	$1DL+1PT+1QT+1CDL+1H+0.45WZ$
160	$1DL+1PT+1QT+1CDL+1H-0.45WZ$
161	$1DL+1PE+1QE+1CDL+1H+1B$

Kombinasi beban terfaktor (*factored load combination* /LRFD) untuk desain rasio *steel*, sambungan dan tulangan beton adalah sebagai berikut:

L/C	KOMBINASI PEMBEBANAN
501	$1.4DL+1.4PE+1.4QE+1.4CDL+1.6H$
502	$0.9DL+0.9PE+0.9QE+0.9CDL+1.6H+1WX$
503	$0.9DL+0.9PE+0.9QE+0.9CDL+1.6H-1WX$
504	$0.9DL+0.9PE+0.9QE+0.9CDL+1.6H+1WZ$
505	$0.9DL+0.9PE+0.9QE+0.9CDL+1.6H-1WZ$
506	$(1.2+0.2Sds)DL+(1.2+0.2Sds)PE+(1.2+0.2Sds)QE+(1.2+0.2Sds)CDL+1.6H+1.3VX+0.39VZ$
507	$(1.2+0.2Sds)DL+(1.2+0.2Sds)PE+(1.2+0.2Sds)QE+(1.2+0.2Sds)CDL+1.6H+1.3VX-0.39VZ$
508	$(1.2+0.2Sds)DL+(1.2+0.2Sds)PE+(1.2+0.2Sds)QE+(1.2+0.2Sds)CDL+1.6H-1.3VX+0.39VZ$
509	$(1.2+0.2Sds)DL+(1.2+0.2Sds)PE+(1.2+0.2Sds)QE+(1.2+0.2Sds)CDL+1.6H-1.3VX-0.39VZ$

510	(1.2 + 0.2 Sds)DL+(1.2 + 0.2 Sds)PE+(1.2 + 0.2 Sds)QE+(1.2 + 0.2 Sds)CDL+1.6H+0.39VX+1.3VZ
511	(1.2 + 0.2 Sds)DL+(1.2 + 0.2 Sds)PE+(1.2 + 0.2 Sds)QE+(1.2 + 0.2 Sds)CDL+1.6H+0.39VX-1.3VZ
512	(1.2 + 0.2 Sds)DL+(1.2 + 0.2 Sds)PE+(1.2 + 0.2 Sds)QE+(1.2 + 0.2 Sds)CDL+1.6H-0.39VX+1.3VZ
513	(1.2 + 0.2 Sds)DL+(1.2 + 0.2 Sds)PE+(1.2 + 0.2 Sds)QE+(1.2 + 0.2 Sds)CDL+1.6H-0.39VX-1.3VZ
514	(0.9 - 0.2 Sds)DL+(0.9 - 0.2 Sds)PE+(0.9 - 0.2 Sds)QE+(0.9 - 0.2 Sds)CDL+1.6H+1.3VX+0.39VZ
515	(0.9 - 0.2 Sds)DL+(0.9 - 0.2 Sds)PE+(0.9 - 0.2 Sds)QE+(0.9 - 0.2 Sds)CDL+1.6H+1.3VX-0.39VZ
516	(0.9 - 0.2 Sds)DL+(0.9 - 0.2 Sds)PE+(0.9 - 0.2 Sds)QE+(0.9 - 0.2 Sds)CDL+1.6H-13VX+0.39VZ
517	(0.9 - 0.2 Sds)DL+(0.9 - 0.2 Sds)PE+(0.9 - 0.2 Sds)QE+(0.9 - 0.2 Sds)CDL+1.6H-13VX-0.39VZ
518	(0.9 - 0.2 Sds)DL+(0.9 - 0.2 Sds)PE+(0.9 - 0.2 Sds)QE+(0.9 - 0.2 Sds)CDL+1.6H+0.39VX+1.3VZ
519	(0.9 - 0.2 Sds)DL+(0.9 - 0.2 Sds)PE+(0.9 - 0.2 Sds)QE+(0.9 - 0.2 Sds)CDL+1.6H+0.39VX-1.3VZ
520	(0.9 - 0.2 Sds)DL+(0.9 - 0.2 Sds)PE+(0.9 - 0.2 Sds)QE+(0.9 - 0.2 Sds)CDL+1.6H-0.39VX+1.3VZ
521	(0.9 - 0.2 Sds)DL+(0.9 - 0.2 Sds)PE+(0.9 - 0.2 Sds)QE+(0.9 - 0.2 Sds)CDL+1.6H-0.39VX-1.3VZ
522	1.4DL+1.4PO+1.4QO+1.4TE+1.4TF+1.4CDL+1.6H
523	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+1.6LL+0.5LR+1.2CDL+1.6CLL+1.6I+1.6H
524	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+1LL+1.6LR+1.2CDL+1CLL+1I+1.6H
525	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+1.6LR+1.2CDL+1CLL+1I+1.6H+0.5WX
526	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+1.6LR+1.2CDL+1CLL+1I+1.6H-0.5WX
527	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+1.6LR+1.2CDL+1CLL+1I+1.6H+0.5WZ
528	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+1.6LR+1.2CDL+1CLL+1I+1.6H-0.5WZ
529	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+0.5LR+1.2CDL+1CLL+1I+1.6H+1WX
530	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+0.5LR+1.2CDL+1CLL+1I+1.6H-1WX
531	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+0.5LR+1.2CDL+1CLL+1I+1.6H+1WZ
532	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+0.5LR+1.2CDL+1CLL+1I+1.6H-1WZ
533	0.9DL+0.9PO+0.9QO+0.9TE+0.9TF+0.9CDL+1.6H+1WX
534	0.9DL+0.9PO+0.9QO+0.9TE+0.9TF+0.9CDL+1.6H-1WX
535	0.9DL+0.9PO+0.9QO+0.9TE+0.9TF+0.9CDL+1.6H+1WZ
536	0.9DL+0.9PO+0.9QO+0.9TE+0.9TF+0.9CDL+1.6H-1WZ
537	(1.2 + 0.2 Sds)DL+(1.2 + 0.2 Sds)PO+(1.2 + 0.2 Sds)QO+(1.2 + 0.2 Sds)TE+(1.2 + 0.2 Sds)TF+1LL+(1.2 + 0.2 Sds)CDL+1CLL+1I+1.6H+1.3VX+0.39VZ
538	(1.2 + 0.2 Sds)DL+(1.2 + 0.2 Sds)PO+(1.2 + 0.2 Sds)QO+(1.2 + 0.2 Sds)TE+(1.2 + 0.2 Sds)TF+1LL+(1.2 + 0.2 Sds)CDL+1CLL+1I+1.6H+1.3VX-0.39VZ
539	(1.2 + 0.2 Sds)DL+(1.2 + 0.2 Sds)PO+(1.2 + 0.2 Sds)QO+(1.2 + 0.2 Sds)TE+(1.2 + 0.2 Sds)TF+1LL+(1.2 + 0.2 Sds)CDL+1CLL+1I+1.6H-1.3VX+0.39VZ
540	(1.2 + 0.2 Sds)DL+(1.2 + 0.2 Sds)PO+(1.2 + 0.2 Sds)QO+(1.2 + 0.2 Sds)TE+(1.2 + 0.2 Sds)TF+1LL+(1.2 + 0.2 Sds)CDL+1CLL+1I+1.6H-1.3VX-0.39VZ
541	(1.2 + 0.2 Sds)DL+(1.2 + 0.2 Sds)PO+(1.2 + 0.2 Sds)QO+(1.2 + 0.2 Sds)TE+(1.2 + 0.2 Sds)TF+1LL+(1.2 + 0.2 Sds)CDL+1CLL+1I+1.6H+0.39VX+1.3VZ
542	(1.2 + 0.2 Sds)DL+(1.2 + 0.2 Sds)PO+(1.2 + 0.2 Sds)QO+(1.2 + 0.2 Sds)TE+(1.2 + 0.2 Sds)TF+1LL+(1.2 + 0.2 Sds)CDL+1CLL+1I+1.6H+0.39VX-1.3VZ
543	(1.2 + 0.2 Sds)DL+(1.2 + 0.2 Sds)PO+(1.2 + 0.2 Sds)QO+(1.2 + 0.2 Sds)TE+(1.2 + 0.2 Sds)TF+1LL+(1.2 + 0.2 Sds)CDL+1CLL+1I+1.6H-0.39VX+1.3VZ
544	(1.2 + 0.2 Sds)DL+(1.2 + 0.2 Sds)PO+(1.2 + 0.2 Sds)QO+(1.2 + 0.2 Sds)TE+(1.2 + 0.2 Sds)TF+1LL+(1.2 + 0.2 Sds)CDL+1CLL+1I+1.6H-0.39VX-1.3VZ
545	(0.9 - 0.2 Sds)DL+(0.9 - 0.2 Sds)PO+(0.9 - 0.2 Sds)QO+(0.9 - 0.2 Sds)TE+(0.9 - 0.2 Sds)TF+(0.9 - 0.2 Sds)CDL+1.6H+1.3VX+0.39VZ
546	(0.9 - 0.2 Sds)DL+(0.9 - 0.2 Sds)PO+(0.9 - 0.2 Sds)QO+(0.9 - 0.2 Sds)TE+(0.9 - 0.2 Sds)TF+(0.9 - 0.2 Sds)CDL+1.6H+1.3VX-0.39VZ
547	(0.9 - 0.2 Sds)DL+(0.9 - 0.2 Sds)PO+(0.9 - 0.2 Sds)QO+(0.9 - 0.2 Sds)TE+(0.9 - 0.2 Sds)TF+(0.9 - 0.2 Sds)CDL+1.6H-1.3VX+0.39VZ
548	(0.9 - 0.2 Sds)DL+(0.9 - 0.2 Sds)PO+(0.9 - 0.2 Sds)QO+(0.9 - 0.2 Sds)TE+(0.9 - 0.2 Sds)TF+(0.9 - 0.2 Sds)CDL+1.6H-1.3VX-0.39VZ

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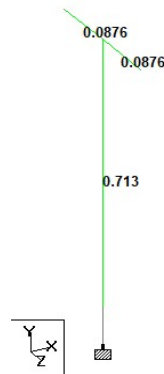
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549	(0.9 - 0.2 Sds)DL+(0.9 - 0.2 Sds)PO+(0.9 - 0.2 Sds)QO+(0.9 - 0.2 Sds)TE+(0.9 - 0.2 Sds)TF+(0.9 - 0.2 Sds)CDL+1.6H+0.39VX+1.3VZ
550	(0.9 - 0.2 Sds)DL+(0.9 - 0.2 Sds)PO+(0.9 - 0.2 Sds)QO+(0.9 - 0.2 Sds)TE+(0.9 - 0.2 Sds)TF+(0.9 - 0.2 Sds)CDL+1.6H+0.39VX-1.3VZ
551	(0.9 - 0.2 Sds)DL+(0.9 - 0.2 Sds)PO+(0.9 - 0.2 Sds)QO+(0.9 - 0.2 Sds)TE+(0.9 - 0.2 Sds)TF+(0.9 - 0.2 Sds)CDL+1.6H-0.39VX+1.3VZ
552	(0.9 - 0.2 Sds)DL+(0.9 - 0.2 Sds)PO+(0.9 - 0.2 Sds)QO+(0.9 - 0.2 Sds)TE+(0.9 - 0.2 Sds)TF+(0.9 - 0.2 Sds)CDL+1.6H-0.39VX-1.3VZ
553	1.2DL+1.2PT+1.2QT+1.2TE+1.2TF+1.6LR+1.2CDL+1.6H+0.5WX
554	1.2DL+1.2PT+1.2QT+1.2TE+1.2TF+1.6LR+1.2CDL+1.6H-0.5WX
555	1.2DL+1.2PT+1.2QT+1.2TE+1.2TF+1.6LR+1.2CDL+1.6H+0.5WZ
556	1.2DL+1.2PT+1.2QT+1.2TE+1.2TF+1.6LR+1.2CDL+1.6H-0.5WZ
557	1.2DL+1.2PT+1.2QT+1.2TE+1.2TF+1.6LR+1.6H
558	0.9DL+0.9PE+0.9QE+0.9CDL+1.6H+1.3B

27.0 HASIL ANALISIS STAAD FONDASI DANGKAL - PIPE SUPPORT TYPE-1C

27.1 Rasio Maksimum Baja Struktural

No.	Member Size	Member Number	L/C	Maximum Ratio	Status	Member Assignment
1	PIPS60	8	522	0.713	OK	Kolom
2	PIPS60	5	522	0.088	OK	Beam



Member Steel Ratio

27.2 Vertical Deflection

Sesuai dengan PHR-SP-CI-GG-002, defleksi yang diizinkan untuk beberapa jenis balok adalah sebagai berikut:
dan untuk balok kantilever = $L/200$.

Design Properties	L/C	Beam No.	Direction	L mm	dv_{all} mm	dv mm	Status	Member Assignment
PIPS60	123	7	Y	500.00	2.50	1.32	OK	Beam

27.3 Horizontal Displacement

- Horizontal Displacement Akibat Beban Angin

Defleksi horizontal yang diizinkan akibat beban angin adalah $H/200$

(ref: PIP STC0105)

Deflection Direction	H mm	dh_{all} mm	Node No.	LC	dhc mm	Status	Member Assignment
X	2500.00	12.50	5	119	11.38	OK	Kolom
Z	2500.00	12.50	5	121	9.30	OK	Kolom

- Horizontal Displacement Akibat Beban Gempa

Defleksi horizontal yang diizinkan akibat beban gempa untuk struktur lain, berdasarkan Tabel 12.12-1 ASCE 7-16, adalah 0,015H untuk Kategori Risiko III

Deflection Direction	H mm	dh _{all} mm	Node No.	LC	dhc mm	Cd	I	dhc. Cd / I mm	Status	Member Assignment
X	2500.00	37.5	5	123	12.68	2.50	1.25	25.36	OK	Kolom
Z	2500.00	37.5	5	127	9.17	2.50	1.25	18.33	OK	Kolom

28.0 ANALISIS FONDASI DANGKAL - PIPE SUPPORT TYPE-1C**28.1 Perhitungan Kapasitas Daya Dukung Fondasi Dangkal**Data Tanah

Sudut geser tanah,	$\phi'_s =$	30.00 °	(Perbaikan tanah)
Kohesi tanah,	$c'_s =$	0.00 kN/m ²	(Perbaikan tanah)
Berat jenis tanah,	$\gamma_s =$	19.00 kN/m ³	(Perbaikan tanah)

Spesifikasi Fondasi

Tipe fondasi,	=	Square
Panjang fondasi,	$L_f =$	1.50 m
Lebar fondasi,	$B_f =$	1.50 m
Kedalaman fondasi dari permukaan tanah,	$D_f =$	0.30 m

Kapasitas Daya Dukung Ultimit Terzaghi untuk Fondasi Dangkal

Perhitungan kapasitas daya dukung pondasi dangkal dilakukan dengan menggunakan rumus keruntuhan geser umum (General Shear Failure) menurut Terzaghi, mengacu pada Principles of Geotechnical Engineering, 7th Edition (Braja M. Das, 2010). Berikut merupakan tahapan perhitungan kapasitas daya dukung pondasi dangkal.

$$q_{u,s} = c'_s \cdot N_c \cdot \xi_c + q \cdot N_q \cdot \xi_q + 0.5 \cdot \gamma_s \cdot B_f \cdot N_\gamma \cdot \xi_\gamma$$

Kontribusi Kohesi Tanah

Koefisien kohesi tanah,	$N_c = \frac{228 + 4.3 \phi'_s}{40 - \phi'_s}$	[Krizek, 1965]
	= 35.70	
Faktor bentuk kohesi tanah,	$\xi_c =$	1.30

Kontribusi Beban Timbunan di Atas Pondasi (Surcharge)

Beban timbunan di atas Footing,	$q = (\gamma_s - \gamma_w) \cdot D_f$	
	= 2.76 kN/m ²	
Koefisien surcharge,	$N_q = \frac{40 + 5 \phi'_s}{40 - \phi'_s}$	[Krizek, 1965]
	= 19.00	
Faktor bentuk surcharge,	$\xi_q =$	1.00

Kontribusi Berat Isi Tanah

Koefisien berat isi tanah,	$N_\gamma = \frac{6 \phi'_s}{40 - \phi'_s}$	[Krizek, 1965]
	= 18.00	
Faktor bentuk berat isi tanah,	$\xi_\gamma =$	0.80
Kapasitas daya dukung ultimit terzaghi,	$q_{u,s} =$	257.58 kN/m ²

Kapasitas Daya Dukung Izin Pondasi Dangkal

Daya dukung ultimit terzaghi, $q_{u,s} = 257.58 \text{ kN/m}^2$

Faktor keamanan daya dukung fondasi, $SF_{bc,f} = 3.00$

Daya dukung izin fondasi, $q_{all,s} = \frac{q_{u,s}}{SF_{bc,f}} = 85.86 \text{ kN/m}^2$

28.2 Reaksi Tumpuan

Beban desain dan beban stabilitas untuk pemeriksaan fondasi disajikan pada tabel berikut.

Kombinasi Beban ASD pada Footing

LC	Kombinasi Beban	Beban					
		F_x	F_y	F_z	M_x	M_y	M_z
		(kN)	(kN)	(kN)	(kN.m)	(kN.m)	(kN.m)
101	1DL+1PE+1QE+1CDL+1H	0.000	37.081	0.000	0.000	0.000	0.000
102	1DL+1PE+1QE+1CDL+1H+0.6WX	-0.115	37.081	0.000	0.000	0.000	0.224
103	1DL+1PE+1QE+1CDL+1H-0.6WX	0.115	37.081	0.000	0.000	0.000	-0.224
104	1DL+1PE+1QE+1CDL+1H+0.6WZ	0.000	37.081	-0.847	-2.200	0.000	0.000
105	1DL+1PE+1QE+1CDL+1H-0.6WZ	0.000	37.081	0.847	2.200	0.000	0.000
106	1.068DL+1.068PE+1.068QE+1.068CDL+1H+0.91VX+0.27VZ	-0.642	39.602	-0.190	-0.390	0.000	1.316
107	1.068DL+1.068PE+1.068QE+1.068CDL+1H+0.91VX-0.27VZ	-0.642	39.602	0.190	0.390	0.000	1.316
108	1.068DL+1.068PE+1.068QE+1.068CDL+1H-0.91VX+0.27VZ	0.642	39.602	-0.190	-0.390	0.000	-1.316
109	1.068DL+1.068PE+1.068QE+1.068CDL+1H-0.91VX-0.27VZ	0.642	39.602	0.190	0.390	0.000	-1.316
110	1.068DL+1.068PE+1.068QE+1.068CDL+1H+0.27VX+0.91VZ	-0.190	39.602	-0.642	-1.316	0.000	0.390
111	1.068DL+1.068PE+1.068QE+1.068CDL+1H+0.27VX-0.91VZ	-0.190	39.602	0.642	1.316	0.000	0.390
112	1.068DL+1.068PE+1.068QE+1.068CDL+1H-0.27VX+0.91VZ	0.190	39.602	-0.642	-1.316	0.000	-0.390
113	1.068DL+1.068PE+1.068QE+1.068CDL+1H-0.27VX-0.91VZ	0.190	39.602	0.642	1.316	0.000	-0.390
114	0.6DL+0.6PE+0.6QE+0.6CDL+1H+0.6WX	-0.115	22.248	0.000	0.000	0.000	0.224
115	0.6DL+0.6PE+0.6QE+0.6CDL+1H-0.6WX	0.115	22.248	0.000	0.000	0.000	-0.224
116	0.6DL+0.6PE+0.6QE+0.6CDL+1H+0.6WZ	0.000	22.248	-0.847	-2.200	0.000	0.000
117	0.6DL+0.6PE+0.6QE+0.6CDL+1H-0.6WZ	0.000	22.248	0.847	2.200	0.000	0.000
118	1DL+1PO+1QO+1TE+1TF+1CDL+1H	-6.780	56.641	-4.800	-12.960	0.000	18.306
119	1DL+1PO+1QO+1TE+1TF+1CDL+1H+0.6WX	-6.895	56.641	-4.800	-12.960	0.000	18.530
120	1DL+1PO+1QO+1TE+1TF+1CDL+1H-0.6WX	-6.665	56.641	-4.800	-12.960	0.000	18.082
121	1DL+1PO+1QO+1TE+1TF+1CDL+1H+0.6WZ	-6.780	56.641	-5.647	-15.160	0.000	18.306
122	1DL+1PO+1QO+1TE+1TF+1CDL+1H-0.6WZ	-6.780	56.641	-3.953	-10.760	0.000	18.306
123	1.068DL+1.068PO+1.068QO+1.068TE+1.068TF+1.068CDL+1H+0.91VX+0.27VZ	-7.883	60.492	-5.317	-14.232	0.000	20.866
124	1.068DL+1.068PO+1.068QO+1.068TE+1.068TF+1.068CDL+1H+0.91VX-0.27VZ	-7.883	60.492	-4.936	-13.451	0.000	20.866
125	1.068DL+1.068PO+1.068QO+1.068TE+1.068TF+1.068CDL+1H-0.91VX+0.27VZ	-6.599	60.492	-5.317	-14.232	0.000	18.235
126	1.068DL+1.068PO+1.068QO+1.068TE+1.068TF+1.068CDL+1H-0.91VX-0.27VZ	-6.599	60.492	-4.936	-13.451	0.000	18.235
127	1.068DL+1.068PO+1.068QO+1.068TE+1.068TF+1.068CDL+1H+0.27VX+0.91VZ	-7.431	60.492	-5.768	-15.157	0.000	19.941
128	1.068DL+1.068PO+1.068QO+1.068TE+1.068TF+1.068CDL+1H+0.27VX-0.91VZ	-7.431	60.492	-4.485	-12.526	0.000	19.941

129	1.068DL+1.068PO+1.068QO+1.068TE+1.068TF+1.068CDL+1H-0.27VX+0.91VZ	-7.051	60.492	-5.768	-15.157	0.000	19.160
130	1.068DL+1.068PO+1.068QO+1.068TE+1.068TF+1.068CDL+1H-0.27VX-0.91VZ	-7.051	60.492	-4.485	-12.526	0.000	19.160
131	1DL+1PO+1QO+1TE+1TF+1LL+1CDL+1CLL+1I+1H	-6.780	56.641	-4.800	-12.960	0.000	18.306
132	1DL+1PO+1QO+1TE+1TF+1LR+1CDL+1CLL+1I+1H	-6.780	56.641	-4.800	-12.960	0.000	18.306
133	1DL+1PO+1QO+1TE+1TF+0.75LL+0.75LR+1CDL+1CLL+1I+1H+0.45WX	-6.866	56.641	-4.800	-12.960	0.000	18.474
134	1DL+1PO+1QO+1TE+1TF+0.75LL+0.75LR+1CDL+1CLL+1I+1H-0.45WX	-6.694	56.641	-4.800	-12.960	0.000	18.138
135	1DL+1PO+1QO+1TE+1TF+0.75LL+0.75LR+1CDL+1CLL+1I+1H+0.45WZ	-6.780	56.641	-5.435	-14.610	0.000	18.306
136	1DL+1PO+1QO+1TE+1TF+0.75LL+0.75LR+1CDL+1CLL+1I+1H-0.45WZ	-6.780	56.641	-4.165	-11.310	0.000	18.306
137	1.05(DL+PO+QO+TE+TF)+0.75LL+1.05CDL+0.75CLL+0.75I+1H+0.68VX+0.2VZ	-7.599	59.473	-5.181	-13.897	0.000	20.204
138	1.05(DL+PO+QO+TE+TF)+0.75LL+1.05CDL+0.75CLL+0.75I+1H+0.68VX-0.2VZ	-7.599	59.473	-4.899	-13.319	0.000	20.204
139	1.05(DL+PO+QO+TE+TF)+0.75LL+1.05CDL+0.75CLL+0.75I+1H-0.68VX+0.2VZ	-6.639	59.473	-5.181	-13.897	0.000	18.238
140	1.05(DL+PO+QO+TE+TF)+0.75LL+1.05CDL+0.75CLL+0.75I+1H-0.68VX-0.2VZ	-6.639	59.473	-4.899	-13.319	0.000	18.238
141	1.05(DL+PO+QO+TE+TF)+0.75LL+1.05CDL+0.75CLL+0.75I+1H+0.2VX+0.68VZ	-7.260	59.473	-5.520	-14.591	0.000	19.510
142	1.05(DL+PO+QO+TE+TF)+0.75LL+1.05CDL+0.75CLL+0.75I+1H+0.2VX-0.68VZ	-7.260	59.473	-4.560	-12.625	0.000	19.510
143	1.05(DL+PO+QO+TE+TF)+0.75LL+1.05CDL+0.75CLL+0.75I+1H-0.2VX+0.68VZ	-6.978	59.473	-5.520	-14.591	0.000	18.932
144	1.05(DL+PO+QO+TE+TF)+0.75LL+1.05CDL+0.75CLL+0.75I+1H-0.2VX-0.68VZ	-6.978	59.473	-4.560	-12.625	0.000	18.932
145	0.6DL+0.6PO+0.6QO+0.6TE+0.6TF+0.6CDL+0.6WX	-4.183	33.984	-2.880	-7.776	0.000	11.208
146	0.6DL+0.6PO+0.6QO+0.6TE+0.6TF+0.6CDL-0.6WX	-3.953	33.984	-2.880	-7.776	0.000	10.760
147	0.6DL+0.6PO+0.6QO+0.6TE+0.6TF+0.6CDL+0.6WZ	-4.068	33.984	-3.727	-9.976	0.000	10.984
148	0.6DL+0.6PO+0.6QO+0.6TE+0.6TF+0.6CDL-0.6WZ	-4.068	33.984	-2.033	-5.576	0.000	10.984
149	0.53DL+0.53PO+0.53QO+0.53TE+0.53TF+0.53CDL+1H+0.91VX+0.27VZ	-4.235	30.020	-2.734	-7.259	0.000	11.018
150	0.53DL+0.53PO+0.53QO+0.53TE+0.53TF+0.53CDL+1H+0.91VX-0.27VZ	-4.235	30.020	-2.354	-6.478	0.000	11.018
151	0.53DL+0.53PO+0.53QO+0.53TE+0.53TF+0.53CDL+1H-0.91VX+0.27VZ	-2.952	30.020	-2.734	-7.259	0.000	8.387
152	0.53DL+0.53PO+0.53QO+0.53TE+0.53TF+0.53CDL+1H-0.91VX-0.27VZ	-2.952	30.020	-2.354	-6.478	0.000	8.387
153	0.53DL+0.53PO+0.53QO+0.53TE+0.53TF+0.53CDL+1H+0.27VX+0.91VZ	-3.784	30.020	-3.186	-8.184	0.000	10.093
154	0.53DL+0.53PO+0.53QO+0.53TE+0.53TF+0.53CDL+1H+0.27VX-0.91VZ	-3.784	30.020	-1.902	-5.553	0.000	10.093
155	0.53DL+0.53PO+0.53QO+0.53TE+0.53TF+0.53CDL+1H-0.27VX+0.91VZ	-3.403	30.020	-3.186	-8.184	0.000	9.312
156	0.53DL+0.53PO+0.53QO+0.53TE+0.53TF+0.53CDL+1H-0.27VX-0.91VZ	-3.403	30.020	-1.902	-5.553	0.000	9.312
157	1DL+1PT+1QT+1CDL+1H+0.45WX	-0.086	58.541	0.000	0.000	0.000	0.168
158	1DL+1PT+1QT+1CDL+1H-0.45WX	0.086	58.541	0.000	0.000	0.000	-0.168

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159	1DL+1PT+1QT+1CDL+1H+0.45WZ	0.000	58.541	-0.635	-1.650	0.000	0.000
160	1DL+1PT+1QT+1CDL+1H-0.45WZ	0.000	58.541	0.635	1.650	0.000	0.000
161	1DL+1PE+1QE+1CDL+1H+1B	0.000	37.081	0.000	0.000	0.000	0.000
NILAI MAKSIMUM		0.642	60.492	0.847	2.200	0.000	20.866
LC MAKSIMUM		108	123	105	105	101	123
NILAI MINIMUM		-7.883	22.248	-5.768	-15.160	0.000	-1.316
LC MINIMUM		123	114	127	121	101	108

Nilai Maksimum dan Minimum Kombinasi Beban ASD pada Footing

Kasus	LC	Kombinasi Beban	Beban					
			F _x	F _y	F _z	M _x	M _y	M _z
			(kN)	(kN)	(kN)	(kN.m)	(kN.m)	(kN.m)
Max F _x	108	1.068DL+1.068PE+1.068QE+1.068CDL+1H-	0.642	39.602	-0.190	-0.390	0.000	-1.316
Min F _x	123	1.068DL+1.068PO+1.068QO+1.068TE+1.068TF+1.068CDL+1H+0.91VX+0.27VZ	-7.883	60.492	-5.317	-14.232	0.000	20.866
Max F _y	123	1.068DL+1.068PO+1.068QO+1.068TE+1.068TF+1.068CDL+1H+0.91VX+0.27VZ	-7.883	60.492	-5.317	-14.232	0.000	20.866
Min F _y	114	0.6DL+0.6PE+0.6QE+0.6CDL+1H+0.6WX	-0.115	22.248	0.000	0.000	0.000	0.224
Max F _z	105	1DL+1PE+1QE+1CDL+1H-0.6WZ	0.000	37.081	0.847	2.200	0.000	0.000
Min F _z	127	1.068DL+1.068PO+1.068QO+1.068TE+1.068TF+1.068CDL+1H+0.27VX+0.91VZ	-7.431	60.492	-5.768	-15.157	0.000	19.941
Max M _x	105	1DL+1PE+1QE+1CDL+1H-0.6WZ	0.000	37.081	0.847	2.200	0.000	0.000
Min M _x	121	1DL+1PO+1QO+1TE+1TF+1CDL+1H+0.6WZ	-6.780	56.641	-5.647	-15.160	0.000	18.306
Max M _y	101	1DL+1PE+1QE+1CDL+1H	0.000	37.081	0.000	0.000	0.000	0.000
Min M _y	101	1DL+1PE+1QE+1CDL+1H	0.000	37.081	0.000	0.000	0.000	0.000
Max M _z	123	1.068DL+1.068PO+1.068QO+1.068TE+1.068TF+1.068CDL+1H+0.91VX+0.27VZ	-7.883	60.492	-5.317	-14.232	0.000	20.866
Min M _z	108	1.068DL+1.068PE+1.068QE+1.068CDL+1H-0.91VX+0.27VZ	0.642	39.602	-0.190	-0.390	0.000	-1.316

Kombinasi Beban LRFD pada Footing

LC	Kombinasi Beban	Beban					
		F _x	F _y	F _z	M _x	M _y	M _z
		(kN)	(kN)	(kN)	(kN.m)	(kN.m)	(kN.m)
501	1.4DL+1.4PE+1.4QE+1.4CDL+1.6H	0.000	51.913	0.000	0.000	0.000	0.000
502	0.9DL+0.9PE+0.9QE+0.9CDL+1.6H+1WX	-0.192	33.373	0.000	0.000	0.000	0.373
503	0.9DL+0.9PE+0.9QE+0.9CDL+1.6H-1WX	0.192	33.373	0.000	0.000	0.000	-0.373
504	0.9DL+0.9PE+0.9QE+0.9CDL+1.6H+1WZ	0.000	33.373	-1.412	-3.667	0.000	0.000
505	0.9DL+0.9PE+0.9QE+0.9CDL+1.6H-1WZ	0.000	33.373	1.412	3.667	0.000	0.000
506	1.3DL+1.3PE+1.3QE+1.3CDL+1.6H+1.3VX+0.39VZ	-0.917	48.205	-0.275	-0.564	0.000	1.880
507	1.3DL+1.3PE+1.3QE+1.3CDL+1.6H+1.3VX-0.39VZ	-0.917	48.205	0.275	0.564	0.000	1.880
508	1.3DL+1.3PE+1.3QE+1.3CDL+1.6H-1.3VX+0.39VZ	0.917	48.205	-0.275	-0.564	0.000	-1.880
509	1.3DL+1.3PE+1.3QE+1.3CDL+1.6H-1.3VX-0.39VZ	0.917	48.205	0.275	0.564	0.000	-1.880
510	1.3DL+1.3PE+1.3QE+1.CDL+1.6H+0.39VX+1.3VZ	-0.275	48.205	-0.917	-1.880	0.000	0.564
511	1.3DL+1.3PE+1.3QE+1.CDL+1.6H+0.39VX-1.3VZ	-0.275	48.205	0.917	1.880	0.000	0.564
512	1.3DL+1.3PE+1.3QE+1.CDL+1.6H-0.39VX+1.3VZ	0.275	48.205	-0.917	-1.880	0.000	-0.564
513	1.3DL+1.3PE+1.3QE+1.CDL+1.6H-0.39VX-1.3VZ	0.275	48.205	0.917	1.880	0.000	-0.564
514	0.8DL+0.8PE+0.8QE+0.8CDL+1.6H+1.3VX+0.39VZ	-0.917	29.664	-0.275	-0.564	0.000	1.880
515	0.8DL+0.8PE+0.8QE+0.8CDL+1.6H+1.3VX-0.39VZ	-0.917	29.664	0.275	0.564	0.000	1.880

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516	0.8DL+0.8PE+0.8QE+0.8CDL+1.6H-1.3VX+0.39VZ	0.917	29.664	-0.275	-0.564	0.000	-1.880
517	0.8DL+0.8PE+0.8QE+0.8CDL+1.6H-1.3VX+0.39VZ	0.917	29.664	0.275	0.564	0.000	-1.880
518	0.8DL+0.8PE+0.8QE+0.8CDL+1.6H+0.39VX+1.3VZ	-0.275	29.664	-0.917	-1.880	0.000	0.564
519	0.8DL+0.8PE+0.8QE+0.8CDL+1.6H+0.39VX-1.3VZ	-0.275	29.664	0.917	1.880	0.000	0.564
520	0.8DL+0.8PE+0.8QE+0.8CDL+1.6H-0.39VX+1.3VZ	0.275	29.664	-0.917	-1.880	0.000	-0.564
521	0.8DL+0.8PE+0.8QE+0.8CDL+1.6H-0.39VX-1.3VZ	0.275	29.664	0.917	1.880	0.000	-0.564
522	1.4DL+1.4PO+1.4QO+1.4TE+1.4TF+1.4CDL+1.6H	-9.492	79.297	-6.720	-18.144	0.000	25.628
523	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+1.6LL+0.5LR+1.2CDL+1.6CLL+1.6H+1.6H	-8.136	67.969	-5.760	-15.552	0.000	21.967
524	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+1.6LL+1.6LR+1.2CDL+1.6CLL+1.6H+1.6H	-8.136	67.969	-5.760	-15.552	0.000	21.967
525	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+1.6LR+1.2CDL+1.6CLL+1.6H+0.5WX	-8.232	67.969	-5.760	-15.552	0.000	22.154
526	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+1.6LR+1.2CDL+1.6CLL+1.6H+0.5WX	-8.040	67.969	-5.760	-15.552	0.000	21.781
527	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+1.6LR+1.2CDL+1.6CLL+1.6H+0.5WZ	-8.136	67.969	-6.466	-17.386	0.000	21.967
528	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+1.6LR+1.2CDL+1.6CLL+1.6H+0.5WZ	-8.136	67.969	-5.054	-13.718	0.000	21.967
529	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+0.5LR+1.2CDL+1.6CLL+1.6H+1WX	-8.328	67.969	-5.760	-15.552	0.000	22.340
530	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+0.5LR+1.2CDL+1.6CLL+1.6H+1WX	-7.944	67.969	-5.760	-15.552	0.000	21.594
531	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+0.5LR+1.2CDL+1.6CLL+1.6H+1WZ	-8.136	67.969	-7.172	-19.219	0.000	21.967
532	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+0.5LR+1.2CDL+1.6CLL+1.6H+1WZ	-8.136	67.969	-4.348	-11.885	0.000	21.967
533	0.9DL+0.9PO+0.9QO+0.9TE+0.9TF+0.9CDL+1.6H+1WX	-6.294	50.977	-4.320	-11.664	0.000	16.849
534	0.9DL+0.9PO+0.9QO+0.9TE+0.9TF+0.9CDL+1.6H+1WX	-5.910	50.977	-4.320	-11.664	0.000	16.102
535	0.9DL+0.9PO+0.9QO+0.9TE+0.9TF+0.9CDL+1.6H+1WZ	-6.102	50.977	-5.732	-15.331	0.000	16.475
536	0.9DL+0.9PO+0.9QO+0.9TE+0.9TF+0.9CDL+1.6H+1WZ	-6.102	50.977	-2.908	-7.997	0.000	16.475
537	1.3(DL+PO+QO+TE+TF)+1LL+1.3CDL+1.6CLL+1.6H+1.3VX+0.39VZ	-9.731	73.633	-6.515	-17.412	0.000	25.677
538	1.3(DL+PO+QO+TE+TF)+1LL+1.3CDL+1.6CLL+1.6H+1.3VX-0.39VZ	-9.731	73.633	-5.965	-16.284	0.000	25.677
539	1.3(DL+PO+QO+TE+TF)+1LL+1.3CDL+1.6CLL+1.6H-1.3VX+0.39VZ	-7.897	73.633	-6.515	-17.412	0.000	21.918
540	1.3(DL+PO+QO+TE+TF)+1LL+1.3CDL+1.6CLL+1.6H-1.3VX-0.39VZ	-7.897	73.633	-5.965	-16.284	0.000	21.918
541	1.3(DL+PO+QO+TE+TF)+1LL+1.3CDL+1.6CLL+1.6H+0.39VX+1.3VZ	-9.089	73.633	-7.157	-18.728	0.000	24.362
542	1.3(DL+PO+QO+TE+TF)+1LL+1.3CDL+1.6CLL+1.6H+0.39VX-1.3VZ	-9.089	73.633	-5.323	-14.968	0.000	24.362
543	1.3(DL+PO+QO+TE+TF)+1LL+1.3CDL+1.6CLL+1.6H-0.39VX+1.3VZ	-8.539	73.633	-7.157	-18.728	0.000	23.234
544	1.3(DL+PO+QO+TE+TF)+1LL+1.3CDL+1.6CLL+1.6H-0.39VX-1.3VZ	-8.539	73.633	-5.323	-14.968	0.000	23.234
545	0.8DL+0.8PO+0.8QO+0.8TE+0.8TF+0.8CDL+1.6H+1.3VX+0.39VZ	-6.341	45.312	-4.115	-10.932	0.000	16.524
546	0.8DL+0.8PO+0.8QO+0.8TE+0.8TF+0.8CDL+1.6H+1.3VX-0.39VZ	-6.341	45.312	-3.565	-9.804	0.000	16.524

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547	0.8DL+0.8PO+0.8QO+0.8TE+0.8TF+0.8CDL+1.6H-1.3VX+0.39VZ	-4.507	45.312	-4.115	-10.932	0.000	12.765
548	0.8DL+0.8PO+0.8QO+0.8TE+0.8TF+0.8CDL+1.6H-1.3VX-0.39VZ	-4.507	45.312	-3.565	-9.804	0.000	12.765
549	0.8DL+0.8PO+0.8QO+0.8TE+0.8TF+0.8CDL+1.6H+0.39VX+1.3VZ	-5.699	45.312	-4.757	-12.248	0.000	15.209
550	0.8DL+0.8PO+0.8QO+0.8TE+0.8TF+0.8CDL+1.6H+0.39VX-1.3VZ	-5.699	45.312	-2.923	-8.488	0.000	15.209
551	0.8DL+0.8PO+0.8QO+0.8TE+0.8TF+0.8CDL+1.6H-0.39VX+1.3VZ	-5.149	45.312	-4.757	-12.248	0.000	14.081
552	0.8DL+0.8PO+0.8QO+0.8TE+0.8TF+0.8CDL+1.6H-0.39VX-1.3VZ	-5.149	45.312	-2.923	-8.488	0.000	14.081
553	1.2DL+1.2PT+1.2QT+1.2TE+1.2TF+1.6LR+1.2CDL+1.6H+0.5WX	-8.232	70.249	-5.760	-15.552	0.000	22.154
554	1.2DL+1.2PT+1.2QT+1.2TE+1.2TF+1.6LR+1.2CDL+1.6H-0.5WX	-8.040	70.249	-5.760	-15.552	0.000	21.781
555	1.2DL+1.2PT+1.2QT+1.2TE+1.2TF+1.6LR+1.2CDL+1.6H+0.5WZ	-8.136	70.249	-6.466	-17.386	0.000	21.967
556	1.2DL+1.2PT+1.2QT+1.2TE+1.2TF+1.6LR+1.2CDL+1.6H-0.5WZ	-8.136	70.249	-5.054	-13.718	0.000	21.967
557	1.2DL+1.2PT+1.2QT+1.2TE+1.2TF+1.6LR+1.6H	-8.136	70.249	-5.760	-15.552	0.000	21.967
558	0.9DL+0.9PE+0.9QE+0.9CDL+1.6H+1.3B	0.000	33.373	0.000	0.000	0.000	0.000
NILAI MAKSIMUM		0.917	79.297	1.412	3.667	0.000	25.677
LC MAKSIMUM		508	522	505	505	501	537
NILAI MINIMUM		-9.731	29.664	-7.172	-19.219	0.000	-1.880
LC MINIMUM		537	514	531	531	501	508

Nilai Maksimum dan Minimum Kombinasi Beban LRFD pada Footing

Kasus Beban	LC	Kombinasi Beban	Beban					
			F_x	F_y	F_z	M_x	M_y	M_z
			(kN)	(kN)	(kN)	(kN.m)	(kN.m)	(kN.m)
Max F_x	508	1.3DL+1.3PE+1.3QE+1.3CDL+1.6H-1.3VX+0.39VZ	0.917	48.205	-0.275	-0.564	0.000	-1.880
Min F_x	537	1.3(DL+PO+QO+TE+TF)+1LL+1.3CDL+1CLL+1I+1.6H+1.3VX+0.39VZ	-9.731	73.633	-6.515	-17.412	0.000	25.677
Max F_y	522	1.4DL+1.4PO+1.4QO+1.4TE+1.4TF+1.4CDL+1.6H	-9.492	79.297	-6.720	-18.144	0.000	25.628
Min F_y	514	0.8DL+0.8PE+0.8QE+0.8CDL+1.6H+1.3VX+0.39VZ	-0.917	29.664	-0.275	-0.564	0.000	1.880
Max F_z	505	0.9DL+0.9PE+0.9QE+0.9CDL+1.6H-1WZ	0.000	33.373	1.412	3.667	0.000	0.000
Min F_z	531	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+0.5LR+1.2CDL+1CLL+1I+1.6H+1WZ	-8.136	67.969	-7.172	-19.219	0.000	21.967
Max M_x	505	0.9DL+0.9PE+0.9QE+0.9CDL+1.6H-1WZ	0.000	33.373	1.412	3.667	0.000	0.000
Min M_x	531	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+0.5LR+1.2CDL+1CLL+1I+1.6H+1WZ	-8.136	67.969	-7.172	-19.219	0.000	21.967
Max M_y	501	1.4DL+1.4PE+1.4QE+1.4CDL+1.6H	0.000	51.913	0.000	0.000	0.000	0.000
Min M_y	501	1.4DL+1.4PE+1.4QE+1.4CDL+1.6H	0.000	51.913	0.000	0.000	0.000	0.000
Max M_z	537	1.3(DL+PO+QO+TE+TF)+1LL+1.3CDL+1CLL+1I+1.6H+1.3VX+0.39VZ	-9.731	73.633	-6.515	-17.412	0.000	25.677
Min M_z	508	1.3DL+1.3PE+1.3QE+1.3CDL+1.6H-1.3VX+0.39VZ	0.917	48.205	-0.275	-0.564	0.000	-1.880

28.3 Pemeriksaan Daya Dukung Tanah

Daya Dukung Izin Tanah

Daya Dukung Ultimit (Teori Terzaghi),

$$q_{all,s} = 85.86 \text{ kN/m}^2$$

Parameter Fondasi

Luas penampang footing,	$A_f =$	2.25 m ²
Mod. penampang elastis thdp sumbu X,	$S_{X,f} =$	0.56 m ²
Mod. penampang elastis thdp sumbu Z,	$S_{Z,f} =$	0.56 m ²

Tekanan yang Bekerja pada Tanah akibat Beban

$$\sigma_{max,f} = \frac{F_Y}{A_f} + \frac{M_X}{S_{X,f}} + \frac{M_Z}{S_{Z,f}} \quad \sigma_{min,f} = \frac{F_Y}{A_f} - \frac{M_X}{S_{X,f}} - \frac{M_Z}{S_{Z,f}}$$

Rekapitulasi Pemeriksaan Daya Dukung Tanah

Kasus Beban	LC	Reaksi Tumpuan ASD dari STAAD.Pro			Daya Dukung Izin Tanah	Tegangan Maksimum pada Tanah	Rasio D/C Tegangan Maksimum	Tegangan Minimum pada Tanah	Rasio D/C Tegangan Minimum	Status
		F_Y	M_X	M_Z						
		(-)	(kN)	(kN.m)	(kN/m ²)	(kN/m ²)	(-)	(kN/m ²)	(-)	(-)
Max F_Y	123	60.492	-14.232	20.866	85.86	38.68	0.45	15.09	0.18	OK
Min F_Y	114	22.248	0.000	0.224	85.86	10.29	0.12	9.49	0.11	OK
Max M_X	105	37.081	2.200	0.000	85.86	20.39	0.24	12.57	0.15	OK
Min M_X	121	56.641	-15.160	18.306	85.86	30.77	0.36	19.58	0.23	OK
Max M_Z	123	60.492	-14.232	20.866	85.86	38.68	0.45	15.09	0.18	OK
Min M_Z	108	39.602	-0.390	-1.316	85.86	14.57	0.17	20.63	0.24	OK
Nilai Maksimum					85.86	38.68	0.45	9.49	0.11	OK

Catatan:

(+) Gaya pada F_Y menunjukkan tekan

(-) Gaya pada F_Y menunjukkan tarik

28.4 Pemeriksaan Stabilitas terhadap GulingMomen Penahan Fondasi

$$M_{rX,f} = F_Y \cdot 0.5L_f$$

$$M_{rZ,f} = F_Y \cdot 0.5B_f$$

Parameter Fondasi

Panjang pondasi,	$L_f =$	1.50 m
Lebar pondasi,	$B_f =$	1.50 m

Faktor Keamanan (SF) terhadap Guling

SF terhadap guling (beban permanen / beban sementara), $SF_{op,f} = 2.00$

Rekapitulasi Pemeriksaan Stabilitas terhadap Guling

Kasus Beban	LC	Reaksi Tumpuan ASD dari STAAD.Pro			Momen Penahan Arah X	Rasio $M_{rX,f} / M_X$	Momen Penahan Arah Z	Rasio $M_{rZ,f} / M_Z$	Status
		F_Y	M_X	M_Z					
		(-)	(kN)	(kN.m)	(kN.m)	(-)	(kN.m)	(-)	(-)
Max F_Y	123	60.492	-14.232	20.866	45.37	3.19	45.37	2.17	OK
Min F_Y	114	22.248	0.000	0.224	16.69	99.00	16.69	99.00	OK
Max M_X	105	37.081	2.200	0.000	27.81	99.00	27.81	99.00	OK

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Min M_x	121	56.641	-15.160	18.306	42.48	2.80	42.48	2.32	OK
Max M_z	123	60.492	-14.232	20.866	45.37	3.19	45.37	2.17	OK
Min M_z	108	39.602	-0.390	-1.316	29.70	76.16	29.70	22.57	OK
Nilai Maksimum		60.49	-15.16	-1.32	16.69	2.80	16.69	2.17	OK

Catatan:

(+) Gaya pada F_y menunjukkan tekan

(-) Gaya pada F_y menunjukkan tarik

28.5 Pemeriksaan Stabilitas terhadap GeserGaya Penahan terhadap Geser

$$F_{rX,f} = F_y \cdot \mu_{s,f}$$

$$F_{rZ,f} = F_y \cdot \mu_{s,f}$$

Parameter Fondasi

Koefisien gesekan, $\mu_{s,f} = 0.50$

Faktor Keamanan (SF) terhadap Geser

SF terhadap geser, $SF_{sl,f} = 1.50$

Rekapitulasi Pemeriksaan Stabilitas terhadap Geser

Kasus Beban	LC	Reaksi Tumpuan ASD dari STAAD.Pro			Gaya Penahan Arah X	Rasio $F_{rX,f} / F_x$	Gaya Penahan Arah Z	Rasio $F_{rZ,f} / F_z$	Status
		F_y	F_x	F_z	$F_{rX,f}$		$F_{rZ,f}$		
		(-)	(kN)	(kN)	(kN)	(-)	(kN)	(-)	
Max F_y	123	60.492	-7.883	-5.317	30.25	3.84	30.25	5.69	OK
Min F_y	114	22.248	-0.115	0.000	11.12	99.00	11.12	99.00	OK
Max F_x	108	39.602	0.642	-0.190	19.80	30.84	19.80	104.22	OK
Min F_x	123	60.492	-7.883	-5.317	30.25	3.84	30.25	5.69	OK
Max F_z	105	37.081	0.000	0.847	18.54	99.00	18.54	99.00	OK
Min F_z	127	60.492	-7.431	-5.768	30.25	4.07	30.25	5.24	OK
Nilai Maksimum					11.12	3.84	11.12	5.24	OK

Catatan:

(+) Gaya pada F_y menunjukkan tekan

(-) Gaya pada F_y menunjukkan tarik

28.6 Pemeriksaan Stabilitas terhadap Gaya AngkatGaya Angkat yang Bekerja pada Fondasi

Kedalaman footing thdp permukaan tanah, $D_f = 0.30 \text{ m}$

Berat isi air, $\gamma_w = 9.81 \text{ kN/m}^3$

Luas penampang alas footing, $A_f = 2.25 \text{ m}^2$

Gaya angkat pada fondasi, $F_{db,f} = D_f \cdot \gamma_w \cdot A_f$

$= 6.62 \text{ kN}$

Gaya Penahan terhadap Gaya Angkat

Gaya penahan terhadap gaya angkat (beban permanen), $F_{rbp,f} = F_{y,Df} \quad (\text{Footing} + \text{Pedestal Selfweight})$

$= 25.06 \text{ kN}$

Gaya penahan terhadap gaya angkat $F_{rbt,f} = \text{Min } F_y \text{ ASD Reaction}$

(beban sementara), = 47.30 kN

Faktor Keamanan (SF) terhadap Gaya Angkat – Beban Permanen

SF thdp gaya angkat (beban permanen), $SF_{b,f} = 1.50$

Nilai SF aktual terhadap gaya angkat, $F_{db,f} = F_{rbp,f} / F_{db,f}$
= 3.78

Cek SF thdp gaya angkat (beban permanen), = OK

Faktor Keamanan (SF) terhadap Gaya Angkat – Beban Sementara

SF thdp gaya angkat (beban sementara), $SF_{b,f} = 1.50$

Nilai SF aktual terhadap gaya angkat, $F_{db,f} = F_{rbt,f} / F_{db,f}$
= 7.14

Cek SF thdp gaya angkat (beban sementara), = OK

29.0 DESAIN TULANGAN FONDASI DANGKAL - PIPE SUPPORT TYPE-1C

29.1 Desain Tulangan Pedestal

Detail perencanaan dan desain tulangan pedestal untuk Fondasi Dangkal - Pipe Support Type-1C. Ringkasan kombinasi pembebanan LRFD pada pedestal adalah sebagai berikut.

LC	Kombinasi Beban	Beban					
		F_x	F_y	F_z	M_x	M_y	M_z
		(kN)	(kN)	(kN)	(kN.m)	(kN.m)	(kN.m)
501	1.4DL+1.4PE+1.4QE+1.4CDL+1.6H	0.000	51.913	0.000	0.000	0.000	0.000
502	0.9DL+0.9PE+0.9QE+0.9CDL+1.6H+1WX	-0.192	33.373	0.000	0.000	0.000	0.373
503	0.9DL+0.9PE+0.9QE+0.9CDL+1.6H-1WX	0.192	33.373	0.000	0.000	0.000	-0.373
504	0.9DL+0.9PE+0.9QE+0.9CDL+1.6H+1WZ	0.000	33.373	-1.412	-3.667	0.000	0.000
505	0.9DL+0.9PE+0.9QE+0.9CDL+1.6H-1WZ	0.000	33.373	1.412	3.667	0.000	0.000
506	1.3DL+1.3PE+1.3QE+1.3CDL+1.6H+1.3VX+0.39VZ	-0.917	48.205	-0.275	-0.564	0.000	1.880
507	1.3DL+1.3PE+1.3QE+1.3CDL+1.6H+1.3VX-0.39VZ	-0.917	48.205	0.275	0.564	0.000	1.880
508	1.3DL+1.3PE+1.3QE+1.3CDL+1.6H-1.3VX+0.39VZ	0.917	48.205	-0.275	-0.564	0.000	-1.880
509	1.3DL+1.3PE+1.3QE+1.3CDL+1.6H-1.3VX-0.39VZ	0.917	48.205	0.275	0.564	0.000	-1.880
510	1.3DL+1.3PE+1.3QE+1.3CDL+1.6H+0.39VX+1.3VZ	-0.275	48.205	-0.917	-1.880	0.000	0.564
511	1.3DL+1.3PE+1.3QE+1.3CDL+1.6H+0.39VX-1.3VZ	-0.275	48.205	0.917	1.880	0.000	0.564
512	1.3DL+1.3PE+1.3QE+1.3CDL+1.6H-0.39VX+1.3VZ	0.275	48.205	-0.917	-1.880	0.000	-0.564
513	1.3DL+1.3PE+1.3QE+1.3CDL+1.6H-0.39VX-1.3VZ	0.275	48.205	0.917	1.880	0.000	-0.564
514	0.8DL+0.8PE+0.8QE+0.8CDL+1.6H+1.3VX+0.39VZ	-0.917	29.664	-0.275	-0.564	0.000	1.880
515	0.8DL+0.8PE+0.8QE+0.8CDL+1.6H+1.3VX-0.39VZ	-0.917	29.664	0.275	0.564	0.000	1.880
516	0.8DL+0.8PE+0.8QE+0.8CDL+1.6H-1.3VX+0.39VZ	0.917	29.664	-0.275	-0.564	0.000	-1.880
517	0.8DL+0.8PE+0.8QE+0.8CDL+1.6H-1.3VX-0.39VZ	0.917	29.664	0.275	0.564	0.000	-1.880
518	0.8DL+0.8PE+0.8QE+0.8CDL+1.6H+0.39VX+1.3VZ	-0.275	29.664	-0.917	-1.880	0.000	0.564
519	0.8DL+0.8PE+0.8QE+0.8CDL+1.6H+0.39VX-1.3VZ	-0.275	29.664	0.917	1.880	0.000	0.564
520	0.8DL+0.8PE+0.8QE+0.8CDL+1.6H-0.39VX+1.3VZ	0.275	29.664	-0.917	-1.880	0.000	-0.564
521	0.8DL+0.8PE+0.8QE+0.8CDL+1.6H-0.39VX-1.3VZ	0.275	29.664	0.917	1.880	0.000	-0.564
522	1.4DL+1.4PO+1.4QO+1.4TE+1.4TF+1.4CDL+1.6H	-9.492	79.297	-6.720	-18.144	0.000	25.628
523	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+1.6LL+0.5LR+1.2CDL+1.6CLL+1.6I+1.6H	-8.136	67.969	-5.760	-15.552	0.000	21.967
524	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+1.6LL+1.6LR+1.2CDL+1.6CLL+1.6I+1.6H	-8.136	67.969	-5.760	-15.552	0.000	21.967
525	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+1.6LR+1.2CDL+1.6CLL+1.6I+1.6H+0.5WX	-8.232	67.969	-5.760	-15.552	0.000	22.154
526	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+1.6LR+1.2CDL+1.6CLL+1.6I+1.6H+0.5WX	-8.040	67.969	-5.760	-15.552	0.000	21.781

527	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+1.6LR+1.2CDL+1CLL+1I+1.6H+0.5WZ	-8.136	67.969	-6.466	-17.386	0.000	21.967
528	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+1.6LR+1.2CDL+1CLL+1I+1.6H+0.5WZ	-8.136	67.969	-5.054	-13.718	0.000	21.967
529	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+0.5LR+1.2CDL+1CLL+1I+1.6H+1WX	-8.328	67.969	-5.760	-15.552	0.000	22.340
530	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+0.5LR+1.2CDL+1CLL+1I+1.6H+1WX	-7.944	67.969	-5.760	-15.552	0.000	21.594
531	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+0.5LR+1.2CDL+1CLL+1I+1.6H+1WZ	-8.136	67.969	-7.172	-19.219	0.000	21.967
532	1.2DL+1.2PO+1.2QO+1.2TE+1.2TF+0.5LR+1.2CDL+1CLL+1I+1.6H+1WZ	-8.136	67.969	-4.348	-11.885	0.000	21.967
533	0.9DL+0.9PO+0.9QO+0.9TE+0.9TF+0.9CDL+1.6H+1WX	-6.294	50.977	-4.320	-11.664	0.000	16.849
534	0.9DL+0.9PO+0.9QO+0.9TE+0.9TF+0.9CDL+1.6H+1WX	-5.910	50.977	-4.320	-11.664	0.000	16.102
535	0.9DL+0.9PO+0.9QO+0.9TE+0.9TF+0.9CDL+1.6H+1WZ	-6.102	50.977	-5.732	-15.331	0.000	16.475
536	0.9DL+0.9PO+0.9QO+0.9TE+0.9TF+0.9CDL+1.6H+1WZ	-6.102	50.977	-2.908	-7.997	0.000	16.475
537	1.3(DL+PO+QO+TE+TF)+1LL+1.3CDL+1CLL+1I+1.6H+1.3VX+0.39VZ	-9.731	73.633	-6.515	-17.412	0.000	25.677
538	1.3(DL+PO+QO+TE+TF)+1LL+1.3CDL+1CLL+1I+1.6H+1.3VX+0.39VZ	-9.731	73.633	-5.965	-16.284	0.000	25.677
539	1.3(DL+PO+QO+TE+TF)+1LL+1.3CDL+1CLL+1I+1.6H+1.3VX+0.39VZ	-7.897	73.633	-6.515	-17.412	0.000	21.918
540	1.3(DL+PO+QO+TE+TF)+1LL+1.3CDL+1CLL+1I+1.6H+1.3VX+0.39VZ	-7.897	73.633	-5.965	-16.284	0.000	21.918
541	1.3(DL+PO+QO+TE+TF)+1LL+1.3CDL+1CLL+1I+1.6H+0.39VX+1.3VZ	-9.089	73.633	-7.157	-18.728	0.000	24.362
542	1.3(DL+PO+QO+TE+TF)+1LL+1.3CDL+1CLL+1I+1.6H+0.39VX-1.3VZ	-9.089	73.633	-5.323	-14.968	0.000	24.362
543	1.3(DL+PO+QO+TE+TF)+1LL+1.3CDL+1CLL+1I+1.6H+0.39VX+1.3VZ	-8.539	73.633	-7.157	-18.728	0.000	23.234
544	1.3(DL+PO+QO+TE+TF)+1LL+1.3CDL+1CLL+1I+1.6H+0.39VX-1.3VZ	-8.539	73.633	-5.323	-14.968	0.000	23.234
545	0.8DL+0.8PO+0.8QO+0.8TE+0.8TF+0.8CDL+1.6H+1.3VX+0.39VZ	-6.341	45.312	-4.115	-10.932	0.000	16.524
546	0.8DL+0.8PO+0.8QO+0.8TE+0.8TF+0.8CDL+1.6H+1.3VX+0.39VZ	-6.341	45.312	-3.565	-9.804	0.000	16.524
547	0.8DL+0.8PO+0.8QO+0.8TE+0.8TF+0.8CDL+1.6H+1.3VX+0.39VZ	-4.507	45.312	-4.115	-10.932	0.000	12.765
548	0.8DL+0.8PO+0.8QO+0.8TE+0.8TF+0.8CDL+1.6H+1.3VX-0.39VZ	-4.507	45.312	-3.565	-9.804	0.000	12.765
549	0.8DL+0.8PO+0.8QO+0.8TE+0.8TF+0.8CDL+1.6H+0.39VX+1.3VZ	-5.699	45.312	-4.757	-12.248	0.000	15.209
550	0.8DL+0.8PO+0.8QO+0.8TE+0.8TF+0.8CDL+1.6H+0.39VX-1.3VZ	-5.699	45.312	-2.923	-8.488	0.000	15.209
551	0.8DL+0.8PO+0.8QO+0.8TE+0.8TF+0.8CDL+1.6H+0.39VX+1.3VZ	-5.149	45.312	-4.757	-12.248	0.000	14.081
552	0.8DL+0.8PO+0.8QO+0.8TE+0.8TF+0.8CDL+1.6H+0.39VX-1.3VZ	-5.149	45.312	-2.923	-8.488	0.000	14.081
553	1.2DL+1.2PT+1.2QT+1.2TE+1.2TF+1.6LR+1.2CDL+1.6H+0.5WX	-8.232	70.249	-5.760	-15.552	0.000	22.154
554	1.2DL+1.2PT+1.2QT+1.2TE+1.2TF+1.6LR+1.2CDL+1.6H+0.5WX	-8.040	70.249	-5.760	-15.552	0.000	21.781
555	1.2DL+1.2PT+1.2QT+1.2TE+1.2TF+1.6LR+1.2CDL+1.6H+0.5WZ	-8.136	70.249	-6.466	-17.386	0.000	21.967
556	1.2DL+1.2PT+1.2QT+1.2TE+1.2TF+1.6LR+1.2CDL+1.6H+0.5WZ	-8.136	70.249	-5.054	-13.718	0.000	21.967
557	1.2DL+1.2PT+1.2QT+1.2TE+1.2TF+1LL+1.6LR+1.6H	-8.136	70.249	-5.760	-15.552	0.000	21.967
558	0.9DL+0.9PE+0.9QE+0.9CDL+1.6H+1.3B	0.000	33.373	0.000	0.000	0.000	0.000
NILAI MAKSIMUM		0.917	79.297	1.412	3.667	0.000	25.677
LC MAKSIMUM		508	522	505	505	501	537
NILAI MINIMUM		-9.731	29.664	-7.172	-19.219	0.000	-1.880
LC MINIMUM		537	514	531	531	501	508

Ringkasan Desain Pedestal

Tipe Pipe Support	Properti Penampang Elemen	REBAR CONFIG.	CASE	LC	F _y	M _x	M _z	PMM RATIO	STATUS
	(mm)				(kN)	(kN.m)	(kN.m)	(-)	(-)
Fondasi Dangkal - Pipe Support Type-1C	400 x 400	8D13	Max F _y	522	79.30	-18.14	25.63	0.1294	OK
			Min F _y	514	29.66	-0.56	1.88	0.0314	OK
			Max M _x	505	33.37	3.67	0.00	0.0647	OK
			Min M _x	531	67.97	-19.22	21.97	0.0659	OK
			Max M _z	537	73.63	-17.41	25.68	0.1375	OK
			Min M _z	508	48.21	-0.56	-1.88	0.0102	OK

29.2 Desain Tulangan FootingDaya Dukung Izin Tanah

Daya dukung izin tanah, $q_{all,s} = 85.86 \text{ kN/m}^2$

Momen Ultimit untuk Desain Tulangan Footing

Beban garis ultimit LRFD pada fondasi, $q_{u,f} = 1.4 \cdot q_{all,s} \cdot 1m$
 $= 120.21 \text{ kN/m}$

Spesifikasi Beton

Kuat tekan beton (struktur), $f'_{c,s} = 28.00 \text{ MPa}$

Spesifikasi Tulangan Memanjang Footing

Modulus elastisitas tulangan $E_r = 200000.00 \text{ MPa}$

Mutu tulangan memanjang, $= (\text{BjTP 280})$

Tegangan leleh tulangan memanjang, $f_{y,rl} = 420.00 \text{ MPa}$

Regangan leleh tulangan memanjang, $\varepsilon_{yt,rl} = 0.0021$

Perhitungan Desain Tulangan Footing

Lokasi		Arah X		Arah Z	
Parameter	Simbol	Nilai	Satuan	Nilai	Satuan
A. Dimensi Footing					
Lebar penampang asumsi ($B_f = 1\text{m}$)	B_f	1000.00	mm	1000.00	mm
Tinggi penampang	H_f	300.00	mm	300.00	mm
Selimit beton	c_c	75.00	mm	75.00	mm
Tinggi efektif penampang	d_f	$H_f - c_c - 0.5D_{rl}$	mm	$H_f - c_c - 0.5D_{rl}$	mm
		218.50	mm	218.50	mm
B. Perencanaan Tulangan Fondasi					
Dia. tulangan memanjang yang digunakan	D_{rl}	13.00	mm	13.00	mm
Spasi tul. memanjang yang digunakan	s_{rl}	150.00	mm	150.00	mm
Jumlah tulangan memanjang yang digunakan	n_{rl}	B_f/s_{rl}	mm ²	B_f/s_{rl}	mm ²
		6.67	ea	6.67	ea
Total longitudinal rebar area used	$A_{rl(used)}$	$n_{rl} \cdot 0.25 \cdot \pi \cdot D_{rl}^2$	mm ²	$n_{rl} \cdot 0.25 \cdot \pi \cdot D_{rl}^2$	mm ²
		884.88	mm ²	884.88	mm ²
Minimum longitudinal rebar area 1 For $f_{y,rl} \geq 420 \text{ MPa}$ [SNI 2847:2019 Table 8.6.1.1]	$A_{rl(min1)}$	$\frac{0.0018 \cdot 420}{f_{y,rl}} A_{g,f}$	mm ²	$\frac{0.0018 \cdot 420}{f_{y,rl}} A_{g,f}$	mm ²
		540.00	mm ²	540.00	mm ²
Minimum longitudinal rebar area 2 For $f_{y,rl} \geq 420 \text{ MPa}$ [SNI 2847:2019 Table 8.6.1.1]	$A_{rl(min2)}$	$0.0014 A_{g,f}$	mm ²	$0.0014 A_{g,f}$	mm ²
		420.00	mm ²	420.00	mm ²
Minimum longitudinal rebar area check [SNI 2847:2019 Section 8.6.1.1]	Condition	$A_{rl(used)} \geq A_{rl(min1)}$ $A_{rl(used)} \geq A_{rl(min2)}$		$A_{rl(used)} \geq A_{rl(min1)}$ $A_{rl(used)} \geq A_{rl(min2)}$	
	Status	OK	-	OK	-
Tinggi blok tegangan tekan	a	$\frac{A_{rl(used)} \cdot f_{y,rl}}{0.85 \cdot f'_{c,s} \cdot B_f}$	mm	$\frac{A_{rl(used)} \cdot f_{y,rl}}{0.85 \cdot f'_{c,s} \cdot B_f}$	mm
		15.62	mm	15.62	mm
Tinggi blok tegangan tekan	c	a/β_1	mm	a/β_1	mm
		18.37	mm	18.37	mm

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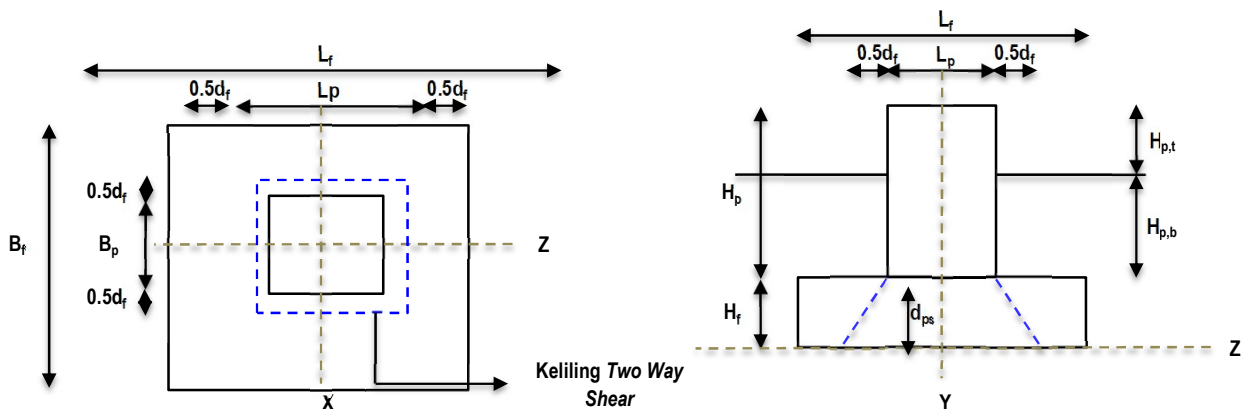
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Regangan aktual tulangan memanjang pada daerah tarik	$\varepsilon_{t,rl}$	$\frac{d_b - c}{c} 0.003$		$\frac{d_b - c}{c} 0.003$	
		0.033	-	0.033	-
Faktor reduksi kekuatan momen [SNI 2847:2019 Table 21.2.2]	ϕ_M	$0.65 \leq 0.65 + \frac{(\varepsilon_{t,rl} - \varepsilon_{y,t,rl})}{(0.005 - \varepsilon_{y,t,rl})} 0.25 \leq 0.90$		$0.65 \leq 0.65 + \frac{(\varepsilon_{t,rl} - \varepsilon_{y,t,rl})}{(0.005 - \varepsilon_{y,t,rl})} 0.25 \leq 0.90$	
		0.90	-	0.90	-
Kapasitas momen footing	$M_{c,f}$	$\phi_M \cdot A_{rl(used)} \cdot f_{y,rl}(d_f - 0.5a)$	kN.m	$\phi_M \cdot A_{rl(used)} \cdot f_{y,rl}(d_f - 0.5a)$	kN.m
		70.47	kN.m	70.47	kN.m
Beban garis ultimit LRFD pada fondasi,	$q_{u,f}$	120.21	kN/m	120.21	kN/m
Panjang beban garis,	$L_{ll,f}$	$0.5B_f$	m	$0.5L_f$	m
		0.75	m	0.75	m
Momen ultimit untuk desain tulangan footing	$M_{u,f}$	$0.5 \cdot q_{u,f} \cdot L_{ll,f}^2$	kN.m	$1/8 \cdot q_{u,f} \cdot L_{ll,f}^2$	kN.m
		33.81	kN.m	8.45	kN.m
Pemeriksaan kapasitas momen footing	Status	OK	-	OK	-

29.3 Pemeriksaan Two Way Shear pada Footing

Two Way Shear Akibat Reaksi Pedestal

Ilustrasi terjadinya geser pons pada fondasi akibat reaksi pedestal ditunjukkan pada gambar di bawah ini.



Ilustrasi Two Way Shear Fondasi Akibat Reaksi Pedestal

Parameter Two Way Shear

Bentuk pedestal,	=	Square
Tipe lokasi pedestal,	=	Corner
Panjang penampang pedestal,	L_p	= 400.00 mm
Lebar penampang pedestal,	B_p	= 400.00 mm
Panjang footing,	L_f	= 1500.00 mm
Lebar footing	B_f	= 1500.00 mm
Tinggi footing	H_f	= 300.00 mm
Tinggi efektif penampang footing	d_f	= 218.50 mm
Tinggi efektif two way shear	$d_{ps} = d_f$	= 218.50 mm
Panjang two way shear	$L_{ps,f} = (L_p + d_f) \leq L_f$	= 618.50 mm
Lebar two way shear	$B_{ps,f} = (B_p + d_f) \leq B_f$	
Keliling two way shear	$b_{0,ps} = 2(L_{ps,f} + B_{ps,f}) \vee \pi L_{ps,f}$	= 1943.08 mm
Rasio panjang pedestal thdp lebar pedestal,	$\beta_p = L_p/B_p$	[SNI 2847:2019 Sec. 22.6.5.2]
	=	1.00

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Koefisien α_s ,	$\alpha_s =$	20.00	[SNI 2847:2019 Sec. 22.6.5.3]
Koefisien modifikasi untuk jenis beton	$\lambda_c =$	1.00	[SNI 2847:2019 Tabel 19.2.4.2]
Kuat tekan beton (beton struktural)	$f'_{c,s} =$	28.00 MPa	

Perhitungan Kapasitas Nominal Two Way Shear Footing

[SNI 2847:2019 Tabel 22.6.5.2]

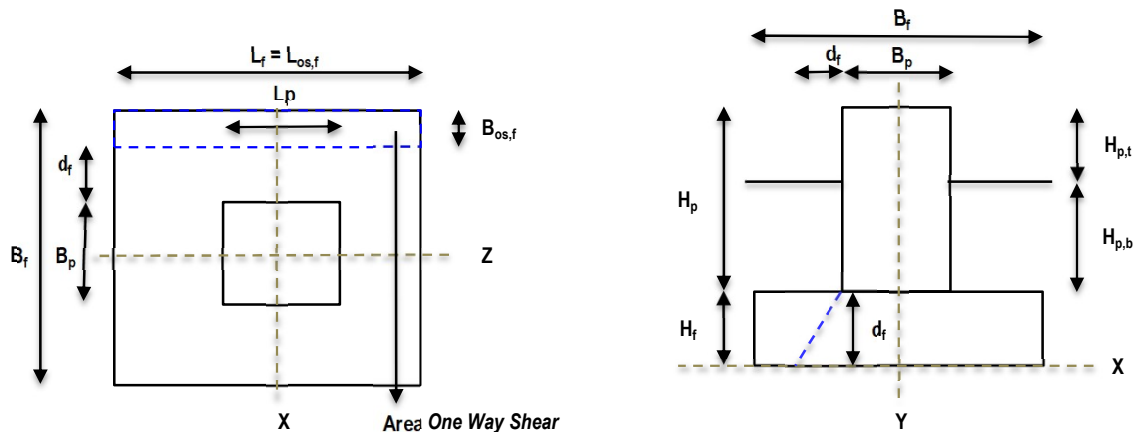
Kapasitas nominal two way shear footing 1,	$V_{nps,f1} =$	$0.33\lambda_c\sqrt{f'_c} \cdot b_{0,ps} \cdot d_{ps}$	
	$=$	741.37 kN	
Kapasitas nominal two way shear footing 1,	$V_{nps,f2} =$	$0.17\left(1 + \frac{2}{\beta_p}\right)\lambda_c \cdot \sqrt{f'_c} \cdot b_{0,ps} \cdot d_{ps}$	
	$=$	1145.75 kN	
Kapasitas nominal two way shear footing 1,	$V_{nps,f3} =$	$0.083\left(2 + \frac{\alpha_s \cdot d_{ps}}{b_{0,ps}}\right)\lambda_c \cdot \sqrt{f'_c} \cdot b_{0,ps} \cdot d_{ps}$	
	$=$	792.29 kN	
Kapasitas nominal two way shear footing yang digunakan,	$V_{nps,f} =$	$\min(V_{nps,f1} ; V_{nps,f2} ; V_{nps,f3})$	
	$=$	741.37 kN	

To Way Shear Check due to Pedestal Reaction

Faktor reduksi kekuatan geser,	$\phi_V =$	0.75	[SNI 2847:2019 Tabel 21.2.1b]
Kapasitas Two Way shear footing tereduksi,	$V_{ps,f} =$	$\phi_V \cdot V_{nps,f}$	
	$=$	556.03 kN	
Beban vertikal ultimit LRFD pada pedestal,	$V_u =$	$qu \cdot (B \cdot L - b_1 \cdot b_2)$	
	$=$	56.19 kN	
Pemeriksaan two way shear ,	$=$	OK	

29.4 Pemeriksaan One Way Shear pada FootingOne Way Shear pada Arah X

Ilustrasi one way shear fondasi pada arah X ditunjukkan pada gambar di bawah ini.



Ilustrasi One Way Shear Fondasi pada Arah X

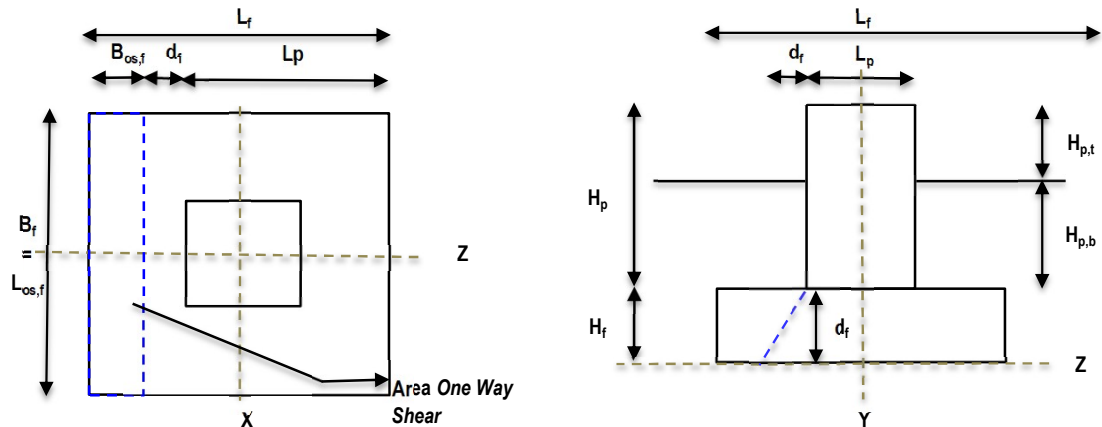
Parameter One Way Shear

Panjang penampang pedestal,	$L_p =$	400.00 mm
Lebar penampang pedestal,	$B_p =$	400.00 mm
Panjang footing ,	$L_f =$	1500.00 mm
Lebar footing	$B_f =$	1500.00 mm
Tinggi footing	$H_f =$	300.00 mm
Tinggi efektif penampang footing	$d_f =$	218.50 mm
Tinggi efektif one way shear,	$d_{os} = d_f$	
	$=$	218.50 mm

Panjang <i>One way shear</i> ,	$L_{os,f} = L_f$	
	$= 1500.00 \text{ mm}$	
Lebar <i>One way shear</i> ,	$B_{os,f} = 0.5B_f - (0.5B_p + d_{os})$	
	$= 331.50 \text{ mm}$	
Luas <i>One way shear</i> ,	$A_{os,f} = L_{os,f} \cdot B_{os,f}$	
	$= 0.50 \text{ m}^2$	
Apakah diperlukan <i>one way shear check</i> ?	$= \text{Diperlukan}$	
Koefisien modifikasi untuk jenis beton,	$\lambda_c = 1.00$	[SNI 2847:2019 Tabel 19.2.4.2]
Kuat tekan beton struktural,	$f'_{c,s} = 28.00 \text{ MPa}$	
<u>Perhitungan Kapasitas Nominal <i>One Way Shear Footing</i></u>		[SNI 2847:2019 Tabel 22.5.5.1]
<i>Footing nominal one way shear capacity</i> ,	$V_{nos,f} = 0.33\lambda_c\sqrt{f'_{c,s}} \cdot L_{os,f} \cdot d_{os}$	
	$= 572.32 \text{ kN}$	
<u>Pemeriksaan <i>One Way Shear</i></u>		
Faktor reduksi kekuatan geser,	$\phi_V = 0.75$	[SNI 2847:2019 Tabel 21.2.1b]
Kapasitas <i>one way shear footing</i> tereduksi,	$V_{os,f} = \phi_V \cdot V_{nos,f}$	
	$= 429.24 \text{ kN}$	
Daya dukung izin tanah,	$q_{all,s} = 85.86 \text{ kN/m}^2$	
Tekanan ultimit LRFD pada fondasi,	$q_{u,f} = 1.4 \cdot q_{all,s}$	
	$= 120.21 \text{ kN/m}^2$	
Beban ultimit LRFD pada fondasi,	$V_u = q_{u,f} \cdot A_{os,f}$	
	$= 59.77 \text{ kN}$	
Hasil pemeriksaan <i>one way shear</i> ,	$= \text{OK}$	

One Way Shear pada Arah Z

Ilustrasi *one way shear* fondasi pada arah Z ditunjukkan pada gambar di bawah ini.



Ilustrasi *One Way Shear* Fondasi pada Arah Z

Parameter *One Way Shear*

Panjang penampang pedestal,	$L_p = 400.00 \text{ mm}$
Lebar penampang pedestal,	$B_p = 400.00 \text{ mm}$
Panjang <i>footing</i> ,	$L_f = 1500.00 \text{ mm}$
Lebar <i>footing</i>	$B_f = 1500.00 \text{ mm}$
Tinggi <i>footing</i>	$H_f = 300.00 \text{ mm}$
Tinggi efektif penampang <i>footing</i>	$d_f = 218.50 \text{ mm}$
Tinggi efektif <i>one way shear</i> ,	$d_{os} = d_f$
	$= 218.50 \text{ mm}$

Panjang <i>One way shear</i> ,	$L_{os,f} = B_f$	
	= 1500.00 mm	
Lebar <i>One way shear</i> ,	$B_{os,f} = 0.5L_f - (0.5L_p + d_{os})$	
	= 331.50 mm	
Luas <i>One way shear</i> ,	$A_{os,f} = L_{os,f} \cdot B_{os,f}$	
	= 0.50 m ²	
Apakah diperlukan <i>one way shear check</i> ?	= Diperlukan	
Koefisien modifikasi untuk jenis beton,	$\lambda_c = 1.00$	[SNI 2847:2019 Table 19.2.4.2]
Kuat tekan beton struktural,	$f'_{c,s} = 28.00$ MPa	
<u>Perhitungan Kapasitas Nominal <i>One Way Shear Footing</i></u>		[SNI 2847:2019 Table 22.5.5.1]
Footing nominal <i>one way shear</i> capacity,	$V_{nos,f} = 0.33\lambda_c\sqrt{f'_{c,s}} \cdot L_{os,f} \cdot d_{os}$	
	= 572.32 kN	
<u>Pemeriksaan <i>One Way Shear</i></u>		
Faktor reduksi kekuatan geser,	$\phi_V = 0.75$	[SNI 2847:2019 Table 21.2.1b]
Kapasitas <i>one way shear footing</i> tereduksi,	$V_{os,f} = \phi_V \cdot V_{nos,f}$	
	= 429.24 kN	
Daya dukung izin tanah,	$q_{all,s} = 85.86$ kN/m ²	
Tekanan ultimit LRFD pada fondasi,	$q_{u,f} = 1.4 \cdot q_{all,s}$	
	= 120.21 kN/m ²	
Beban ultimit LRFD pada fondasi,	$V_u = q_{u,f} \cdot A_{os,f}$	
	= 59.77 kN	
Hasil pemeriksaan <i>one way shear</i> ,	= OK	

30.0 RINGKASAN DESAIN FONDASI DANGKAL - PIPE SUPPORT TYPE-1C

Desain Fondasi Dangkal - Pipe Support Type-1C

Dimensi	Tebal	Tulangan Utama		Elevasi Dasar Footing
		Arah X	Arah Z	
(m)	(m)	(-)	(-)	(m)
1.5 x 1.5	0.300	D13-150	D13-150	-0.500
Status	OK	OK	OK	(-)

Ringkasan Pemeriksaan Stabilitas Fondasi Dangkal - Pipe Support CGS-1 Tipe C

Parameter Stabilitas	Beban		Kapasitas		Rasio Kritis D/C	Status
	Nilai	Satuan	Nilai	Satuan		
Pemeriksaan Daya Dukung Tanah	38.68	kN/m ²	85.86	kN/m ²	0.45	OK ; <1
Stabilitas terhadap Guling Arah X	-15.16	kN.m	16.69	kN.m	2.80	OK ; >2
Stabilitas terhadap Guling Arah Z	-1.32	kN.m	16.69	kN.m	2.17	OK ; >2
Stabilitas terhadap Geser Arah X	7.88	kN	11.12	kN	3.84	OK ; >1.5
Stabilitas terhadap Geser Arah Z	5.77	kN	11.12	kN	5.24	OK ; >1.5
Stabilitas terhadap Gaya Angkat Permanen	6.62	kN	25.06	kN	3.78	OK ; >1.5
Stabilitas terhadap Gaya Angkat Temporer	6.62	kN	47.30	kN	7.14	OK ; >1.5
Punching Shear Akibat Reaksi Pedestal	56.19	kN	556.03	kN	-	OK
One Way Shear Arah X	59.77	kN	429.24	kN	-	OK
One Way Shear Arah Z	59.77	kN	429.24	kN	-	OK

30.1 Cek Penurunan Tanah

Penurunan/ *Settlement* fondasi dibagi menjadi 3 komponen yaitu : Penurunan segera (*immediate settlement*), penurunan konsolidasi primer dan penurunan konsolidasi sekunder.

Penurunan total adalah jumlah dari ketiga komponen penurunan tersebut, atau bila dinyatakan dalam persamaan adalah sbb:

$$S = S_i + S_{c1} + S_{c2}$$

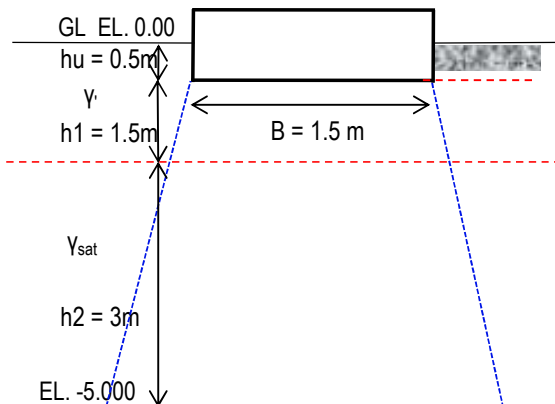
dimana, S = Penurunan Total

S_i = Penurunan Segera

S_{c1} = Penurunan Konsolidasi Primer

S_{c2} = Penurunan Konsolidasi Sekunder

Data tanah



γ_{wet}	=	1089	kg/m ³	(berat jenis tanah basah)
γ_{sat}	=	2070	kg/m ³	(berat jenis tanah jenuh)
γ'	=	$\gamma_{sat} - \gamma_w$	=	1089 kg/m ³
e_o	=	0.500	(asumsi)	
E_s	=	$1000 \cdot (N^3)$	kPa	dengan N = 4
	=	12000	kPa	Sand Gravel (Bowles, 1996)
	=	12232415.9	kg/m ²	
C_c	=	0.12	(compression index)	
C_s	=	$1/10 \times C_c$	=	0.0116 (swelling index)
P_c'	=	1	kg/cm ²	= 10000 kg/m ² (Asumsi)
μ	=	0.3		
z	=	5.00		
h_u	=	0.50	m	(Hfb)
h_1	=	1.50	m	
h_2	=	3.00	m	

Penurunan Segera:

dimana,

S_i : *Immediate settlement*

q_0 : tekanan di dasar fondasi

B : lebar fondasi

μ : *poisson ratio*

γ_s : Berat jenis

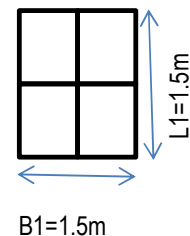
c : Kohesi Tanah

ϕ : Sudut Geser

E_s : *Young Modulus Elasticity*

$$S_i = q_0 B' \frac{1 - \mu^2}{E_s} \left(I_1 + \frac{1 - 2\mu}{1 - \mu} I_2 \right) I_F$$

=	38.68	kN/m ²	L_1	=	1.50
=	1.50	m	B_1	=	1.50
=	0.3				
=	10.89	kN/m ³			
=	0.00	kN/m ²			
=	30.00	°			
=	12000	kN/m ²			



Nilai faktor pengaruh I_1 / I_2 , dapat diperoleh melalui persamaan steinbrenner (1934) sbb:

$$\text{Influence factor : } M = L_1 / B_1 = 1.000$$

$$N = (h_1 + h_2) / B_f = 3.000$$

$$I_1 = \left[M \cdot \ln \left[\frac{(1 + \sqrt{M^2 + 1}) \cdot \sqrt{M^2 + N^2}}{M \cdot (1 + \sqrt{M^2 + N^2 + 1})} \right] + \ln \left[\frac{(M + \sqrt{M^2 + 1}) \cdot \sqrt{1 + N^2}}{(M + \sqrt{M^2 + N^2 + 1})} \right] \right] = 0.363$$

$$I_2 = \frac{\pi}{2} \cdot \text{atan} \left(\frac{M}{N \cdot \sqrt{M^2 + N^2 + 1}} \right) = 0.048$$

Steinbrenner Influence factor (I_s):

$$I_s = I_1 + \frac{(1-2\nu) \cdot I_2}{1-\nu} = 0.390$$

Nilai I_f , diperoleh dari grafik :depth ratio (D/B)

$$\frac{5 \cdot H_{fb}}{B_f} = 3.000$$

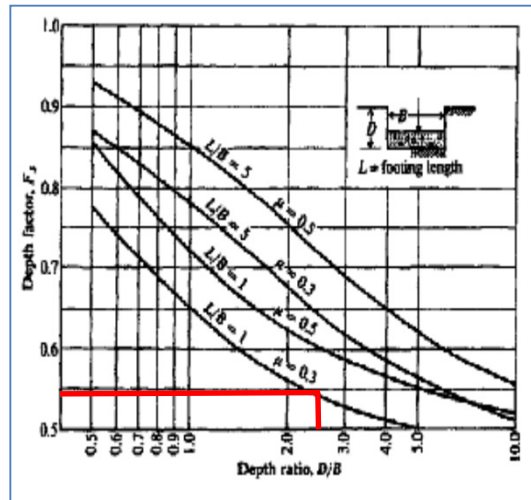
Length ratio (L/B)

$$\frac{L}{B} = 1.000$$

Interpolasi

D/B	I_f
5	0.62
1.00	0.53
1	0.53

$$I_f = \mathbf{0.53}$$



Penurunan segera di tengah fondasi :

$$S_i = 38.68 \times 1.5 \left[\frac{(1-0.3^2)}{12000} \right] \times [0.36 + (1-2(0.3)/(1-0.3)) \times 0.048 \times 0.53 \times 4]$$

$$= 0.003641 \text{ m}$$

$$\mathbf{S_i = 3.641 \text{ mm}}$$

Penurunan Konsolidasi

Penurunan konsolidasi di hitung berdasarkan dengan persamaan yang berbeda tergantung dengan syarat sbb:

Clay normally consolidated (NC)

$$S_c = (C_c \cdot H) / (1+e_0) \log (p_o' + \Delta p) / p_o'$$

Clay over consolidated with $p_o \leq p_c$ (OC)

$$(p_o' + \Delta p > p_c')$$

$$S_c = (C_s \cdot H) / (1+e_0) \log (p_o' + \Delta p) / p_o'$$

Tekanan *overburden* efektif ditengah lapisan lempung

$$p_o' = \gamma' \times h_1 + \gamma_{sat} \times h_2 + \gamma_w \times 0.5 \times h_2$$

$$= 9314.6 \text{ kg/m}^2 < p_c' = 10000 \text{ kg/m}^2$$

karena $p_o' < p_c'$, maka tanah termasuk jenis lempung *overconsolidated*

Tegangan tambahan dibawah fondasi

$$\Delta p = 4.1 \cdot q_0$$

$$= 466.1 \text{ kg/m}^2$$

dimana nilai I didapat dari grafik fadum, 1941 sbb:

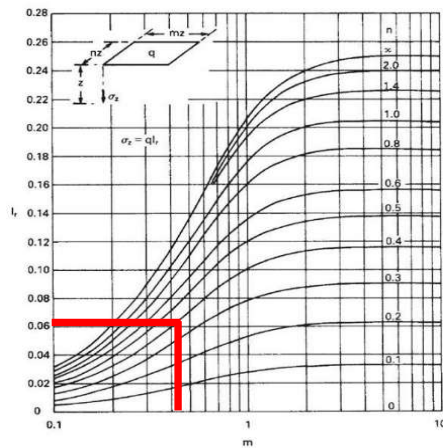


Figure 5.10 Vertical stress under a corner of a rectangular area carrying a uniform pressure.
(Reproduced from R.E. Fadum (1948) *Proceedings of the 2nd International Conference of SMFE*, Rotterdam, Vol. 3, by permission of Professor Fadum.)

$$n = B/Z = 0.30$$

$$m = L/Z = 0.30$$

Interpolasi

m	I
1.00	0.08
0.30	0.03
0.20	0.023

$$I = 0.030$$

$$\text{Karena } p_o' + \Delta p = 466.1 \text{ kg/m}^2 > p_c'$$

maka,

$$\begin{aligned} \Delta e &= [C_s \log (p_c'/p_o')] + [C_c \log ((p_o' + \Delta p)/p_c')] \\ &= -0.000762191 \text{ m} \end{aligned}$$

$$\begin{aligned} S_s &= \frac{\Delta e}{1 + e_0} \cdot H_2 \\ &= -0.00152 \text{ m} \end{aligned}$$

Besar penurunan konsolidasi primer adalah :

$$\begin{aligned} Sc_1 &= (C_s \cdot H)/(1 + e_0) \log (p_o' + \Delta p)/p_o' \quad (H = h_u + h_1) \\ &= 0.00033 \text{ m} \\ &= 0.329 \text{ mm} \end{aligned}$$

Besar penurunan konsolidasi sekunder adalah :

$$\begin{aligned} Sc_2 &= (C_s \cdot Z)/(1 + e_0) \log (p_o' + \Delta p)/p_o' \\ &= 0.00082 \text{ m} \\ &= 0.823 \text{ mm} \end{aligned}$$

Total penurunan fondasi dangkal pada struktur fondasi adalah :

$$\begin{aligned} Stot &= S_i + Sc_1 + Sc_2 \\ &= 4.793 \text{ mm} \end{aligned}$$

Status = Total penurunan fondasi < 25 mm --> OK